NOTICE OF CHANGE

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MIL-HDBK-217F
NOTICE 2
28 February 1995

MILITARY HANDBOOK
RELIABILITY PREDICTION OF ELECTRONIC EQUIPMENT

To all holders of MIL-HDBK-217F

1. The following pages of MIL-HDBK-217F have been revised and supersede the pages listed.

New Page(s)	Date	Superseded Page(s)	Date
Front Cover		Front Cover	2 December 1991
iii	2 December 1991	iii	Reprinted without change
iv		iv	2 December 1991
v		v	2 December 1991
vi		vi	2 December 1991
vii		vii	10 July 1992
viii		New Page	
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1-2		New Page	
2-1		2-1	2 December 1991
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2-3		2-3	2 December 1991
2-4		2-4	2 December 1991
2-5		2-5	2 December 1991
2-6		2-6	2 December 1991
2-7		New Page	
2-8		New Page	
5-3		5-3	10 July 1992
5-4	10 July 1992	5-4	Reprinted without change
5-5		5-5	2 December 1991
5-6		5-6	2 December 1991
5-9		5-9	10 July 1992
5-10	2 December 1991	5-10	Reprinted without change
5-23		5-23	2 December 1991
5-24		5-24	2 December 1991
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6-2		6-2	2 December 1991
7-3		7-3	2 December 1991
7-4	2 December 1991	7-4	Reprinted without change
9-1 through 9-3		9-1 through 9-29	2 December 1991
10-1 through 10-6		10-1 through 10-32	2 December 1991
11-1		11-1	2 December 1991
11-2		11-2	2 December 1991
11-3		11-3	2 December 1991
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12-1		12-1	2 December 1991
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12-4		12-4	2 December 1991
12-5		12-5	2 December 1991
13-1		13-1	2 December 1991
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14-1 through 14-2		14-1 through 14-4	2 December 1991
14-3		14-5	2 December 1991
15-1 through 15-3		15-1 through 15-6	2 December 1991
16-1		16-1	2 December 1991
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17-1		17-1	2 December 1991
Appendix A		A-1 through A-18	2 December 1991, 10 July 1992
C-3		C-3	2 December 1991
C-4		C-4	2 December 1991

2. Retain the pages of this notice and insert before the Table of Contents.

3. Holders of MIL-HDBK-217F will verify that page changes and additions indicated have been entered. The notice pages will be retained as a check sheet. The issuance, together with appended pages, is a separate publication. Each notice is to be retained by stocking points until the military handbook is revised or canceled.

Custodians:
Army - CR
Navy - EC
Air Force - 17

Preparing Activity:
Air Force - 17

Project No. RELI-0074

Review Activities:
Army - MI, AV, ER
Navy - SH, AS, OS
Air Force - 11, 13, 15, 19, 99

Army - AT, ME, GL
Navy - CG, MC, YD, TD
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MIL-HDBK-217F

2 DECEMBER 1991

SUPERSEDING
MIL-HDBK-217E, Notice 1
2 January 1990

MILITARY HANDBOOK

RELIABILITY PREDICTION OF ELECTRONIC EQUIPMENT



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DEPARTMENT OF DEFENSE
WASHINGTON DC 20301

RELIABILITY PREDICTION OF ELECTRONIC EQUIPMENT

1. This standardization handbook was developed by the Department of Defense with the assistance of the military departments, federal agencies, and industry.

2. Every effort has been made to reflect the latest information on reliability prediction procedures. It is the intent to review this handbook periodically to ensure its completeness and currency.

3. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Laboratory/ERSR, Attn: Seymour F. Morris, 525 Brooks Rd., Griffiss AFB, NY 13441-4505, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FOREWORD

1.0 THIS HANDBOOK IS FOR GUIDANCE ONLY. THIS HANDBOOK SHALL NOT BE CITED AS A REQUIREMENT. IF IT IS, THE CONTRACTOR DOES NOT HAVE TO COMPLY.

MIL-HDBK-217F, Notice 2 provides the following changes based upon a recently completed study (see Ref. 37 listed in Appendix C):

- Revised resistor and capacitor models, including new models to address chip devices.
- Updated failure rate models for transformers, coils, motors, relays, switches, circuit breakers, connectors, printed circuit boards (with and without surface mount technology) and connections.
- A new model to address surface mounted technology solder connections.
- A revised Traveling Wave Tube model based upon data supplied by the Electronic Industries Association Microwave Tube Division. This further lowers the calculated failure rates beyond the earlier modifications made in the base document (MIL-HDBK-217F, 2 December 1991).
- Revised the Fast Recovery Power Rectifier base failure rate downward based on a reevaluation of Ref. 28.

2.0 MIL-HDBK-217F, Notice 1, (10 July 1992) was issued to correct minor typographical errors in the basic F Revision.

3.0 MIL-HDBK-217F, (base document), (2 December 1991) provided the following changes based upon recently completed studies (see Ref. 30 and 32 listed in Appendix C):

1. New failure rate prediction models are provided for the following nine major classes of microcircuits:
   - Monolithic Bipolar Digital and Linear Gate/Logic Array Devices
   - Monolithic MOS Digital and Linear Gate/Logic Array Devices
   - Monolithic Bipolar and MOS Digital Microprocessor Devices (including Controllers)
   - Monolithic Bipolar and MOS Memory Devices
   - Monolithic GaAs Digital Devices
   - Monolithic GaAs MMIC Devices
   - Hybrid Microcircuits
   - Magnetic Bubble Memories
   - Surface Acoustic Wave Devices

   The 2 December 1991 revision provided new prediction models for bipolar and MOS microcircuits with gate counts up to 60,000, linear microcircuits with up to 3000 transistors, bipolar and MOS digital microprocessor and co-processors up to 32 bits, memory devices with up to 1 million bits, GaAs monolithic microwave integrated circuits (MMICs) with up to 1,000 active elements, and GaAs digital ICs with up to 10,000 transistors. The C_1 factors have been extensively revised to reflect new technology devices with improved reliability, and the activation energies representing the temperature sensitivity of the dice (π_T) have been changed for MOS devices and for memories. The

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C_2 factor remains unchanged from the previous Handbook version, but includes pin grid arrays and surface mount packages using the same model as hermetic, solder-sealed dual in-line packages. New values have been included for the quality factor (π_Q), the learning factor (π_L), and the environmental factor (π_E). The model for hybrid microcircuits has been revised to be simpler to use, to delete the temperature dependence of the seal and interconnect failure rate contributions, and to provide a method of calculating chip junction temperatures.

2. A new model for Very High Speed Integrated Circuits (VHSIC/VHSIC Like) and Very Large Scale Integration (VLSI) devices (gate counts above 60,000).

3. The reformatting of the entire handbook to make it easier to use.

4. A reduction in the number of environmental factors (π_E) from 27 to 14.

5. A revised failure rate model for Network Resistors.

6. Revised models for TWTs and Klystrons based on data supplied by the Electronic Industries Association Microwave Tube Division.

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1.0 SCOPE

1.1 Purpose - This handbook is for guidance only and shall not be cited as a requirement. If it is, the contractor does not have to comply (see Page 1-2). The purpose of this handbook is to establish and maintain consistent and uniform methods for estimating the inherent reliability (i.e., the reliability of a mature design) of military electronic equipment and systems. It provides a common basis for reliability predictions during acquisition programs for military electronic systems and equipment. It also establishes a common basis for comparing and evaluating reliability predictions of related or competitive designs. The handbook is intended to be used as a tool to increase the reliability of the equipment being designed.

1.2 Application - This handbook contains two methods of reliability prediction - "Part Stress Analysis" in Sections 5 through 23 and "Parts Count" in Appendix A. These methods vary in degree of information needed to apply them. The Part Stress Analysis Method requires a greater amount of detailed information and is applicable during the later design phase when actual hardware and circuits are being designed. The Parts Count Method requires less information, generally part quantities, quality level, and the application environment. This method is applicable during the early design phase and during proposal formulation. In general, the Parts Count Method will usually result in a more conservative estimate (i.e., higher failure rate) of system reliability than the Parts Stress Method.

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OFFICE OF THE ASSISTANT SECRETARY OF DEFENSE
3300 DEFENSE PENTAGON
WASHINGTON, DC 20301-3300

ECONOMIC SECURITY

FEB 28 1995

COMMANDER, ROME LABORATORY (AFMC), ATTN: RL/ERSR, MR. S. MORRIS

SUBJECT: Notice 2 to MIL-HDBK-217F, "Reliability Prediction of Electronic Equipment", Project RELI-0074

Prior to sending the subject notice to the DoD Single Stock Point for printing and distribution, the following additions must be made:

- Across the cover in BIG BOLD BLACK LETTERS - ALL CAPS: Insert "THIS HANDBOOK IS FOR GUIDANCE ONLY. DO NOT CITE THIS DOCUMENT AS A REQUIREMENT".
- In the FOREWORD (Page vii of Notice 2), paragraph 1.0: Add "THIS HANDBOOK IS FOR GUIDANCE ONLY. THIS HANDBOOK SHALL NOT BE CITED AS A REQUIREMENT. IF IT IS, THE CONTRACTOR DOES NOT HAVE TO COMPLY."
- Add an entry for the SCOPE, paragraph 1.1 (Purpose): "This handbook is for guidance only and shall not be cited as a requirement. If it is, the contractor does not have to comply."

If you have any questions regarding this request, please contact Ms. Carla Jenkins.

Walter B. Bergmann, II
Chairman,
Defense Standards Improvement Council

cc: OUSD(A&T)DTSE&E/SE, Mr. M. Zsak

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2.0 REFERENCE DOCUMENTS

This handbook cites some specifications which have been cancelled or which describe devices that are not to be used for new design. This information is necessary because some of these devices are used in so-called "off-the-shelf" equipment which the Department of Defense purchases. The documents cited in this section are for guidance and information.

SPECIFICATION	SECTION #	TITLE
MIL-C-5	10.1	Capacitors, Fixed, Mica Dielectric, General Specification for
MIL-R-11	9.1	Resistor, Fixed, Composition (Insulated), General Specification for
MIL-R-19	9.1	Resistor, Variable, Wirewound (Low Operating Temperature) General Specification for
MIL-C-20	10.1	Capacitor, Fixed, Ceramic Dielectric (Temperature Compensating), Established Reliability and Nonestablished Reliability, General Specification for
MIL-R-22	9.1	Resistor, Variable, Wirewound (Power Type), General Specification for
MIL-C-25	10.1	Capacitor, Fixed, Paper-Dielectric, Direct Current (Hermetically Sealed in Metal Cases), General Specification for
MIL-R-26	9.1	Resistor, Fixed, Wirewound (Power Type), General Specification for
MIL-T-27	11.1	Transformer and Inductors (Audio, Power, High Power Pulse), General Specification for
MIL-C-62	10.1	Capacitor, Fixed Electrolytic (DC, Aluminum, Dry Electrolyte, Polarized), General Specification for
MIL-C-81	10.1	Capacitor, Variable, Ceramic Dielectric, General Specification for
MIL-C-92	10.1	Capacitor, Variable, Air Dielectric (Trimmer), General Specification for
MIL-R-93	9.1	Resistor, Fixed, Wirewound (Accurate), General Specification for
MIL-R-94	9.14	Resistor, Variable, Composition, General Specification for
MIL-V-95	23.1	Vibrator, Interrupter and Self-Rectifying, General Specification for
W-L-111	20.1	Lamp, Incandescent Miniature, Tungsten Filament
W-C-375	14.5	Circuit Breaker, Molded Case, Branch Circuit and Service
W-F-1726	22.1	Fuse, Cartridge, Class H (this covers renewable and nonrenewable)
W-F-1814	22.1	Fuse, Cartridge, High Interrupting Capacity
MIL-C-3098	19.1	Crystal Unit, Quartz, General Specification for
MIL-C-3607	15.1	Connector, Coaxial, Radio Frequency, Series Pulse, General Specifications for
MIL-C-3643	15.1	Connector, Coaxial, Radio Frequency, Series HN and Associated Fittings, General Specification for

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MIL-C-3650	15.1	Connector, Coaxial, Radio Frequency, Series LC
MIL-C-3655	15.1	Connector, Plug and Receptacle, Electrical (Coaxial Series Twin) and Associated Fittings, General Specification for
MIL-S-3786	14.3	Switch, Rotary (Circuit Selector, Low-Current (Capacity)), General Specification for
MIL-S-3950	14.1	Switch, Toggle, Environmentally Sealed, General Specification for
MIL-C-3965	10.1	Capacitor, Fixed, Electrolytic (Nonsolid Electrolyte), Tantalum, General Specification for
MIL-C-5015	15.1	Connector, Electrical, Circular Threaded, AN Type, General Specification for
MIL-F-5372	22.1	Fuse, Current Limiter Type, Aircraft
MIL-S-5594	14.1	Switches, Toggle, Electrically Held Sealed, General Specification for
MIL-R-5757	13.1	Relays, Electromagnetic, General Specification for
MIL-R-6106	13.1	Relay, Electromagnetic (Including Established Reliability (ER) Types), General Specification for
MIL-L-6363	20.1	Lamp, Incandescent, Aircraft Service, General Specification for
MIL-S-8805	14.1, 14.2	Switches and Switch Assemblies, Sensitive and Push (Snap Action), General Specification for
MIL-S-8834	14.1	Switches, Toggle, Positive Break, General Specification for
MIL-S-8932	14.1	Switches, Pressure, Aircraft, General Specification for
MIL-S-9395	14.1	Switches, Pressure, (Absolute, Gage, and Differential), General Specification for
MIL-S-9419	14.1	Switch, Toggle, Momentary Four Position On, Center Off, General Specification for
MIL-M-10304	18.1	Meter, Electrical Indicating, Panel Type, Ruggedized, General Specification for
MIL-R-10509	9.1	Resistor, Fixed Film (High Reliability), General Specification for
MIL-C-10950	10.1	Capacitor, Fixed, Mica Dielectric, Button Style, General Specification for
MIL-C-11015	10.1	Capacitor, Fixed, Ceramic Dielectric (General Purpose), General Specification for
MIL-C-11272	10.1	Capacitor, Fixed, Glass Dielectric, General Specification for

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MIL-C-11693	10.1	Capacitor, Feed Through, Radio Interference Reduction AC and DC, (Hermetically Sealed in Metal Cases) Established and Nonestablished Reliability, General Specification for
MIL-R-11804	9.1	Resistor, Fixed, Film (Power Type), General Specification for
MIL-S-12211	14.1	Switch, Pressure
MIL-S-12285	14.1	Switches, Thermostatic
MIL-S-12883	15.3	Sockets and Accessories for Plug-In Electronic Components, General Specification for
MIL-C-12889	10.1	Capacitor, By-Pass, Radio - Interference Reduction, Paper Dielectric, AC and DC, (Hermetically Sealed in Metallic Cases), General Specification for
MIL-R-12934	9.1	Resistor, Variable, Wirewound, Precision, General Specification for
MIL-S-13484	14.1	Switch, Sensitive: 30 Volts Direct Current Maximum, Waterproof
MIL-C-13516	14.2	Circuit Breakers, Manual and Automatic (28 Volts DC)
MIL-S-13623	14.1	Switches, Rotary: 28 Volt DC
MIL-R-13718	13.1	Relays, Electromagnetic 24 Volt DC
MIL-S-13735	14.1	Switches, Toggle: 28 Volt DC
MIL-C-14409	10.1	Capacitor, Variable (Piston Type, Tubular Trimmer), General Specification for
MIL-F-15160	22.1	Fuse, Instrument, Power and Telephone
MIL-S-15291	14.1	Switches, Rotary, Snap Action and Detent/Spring Return Action, General Specification for
MIL-C-15305	11.2	Coils, Electrical, Fixed and Variable, Radio Frequency, General Specification for
MIL-C-15370	15.1	Couplers, Directional, General Specification for
MIL-F-15733	21.1	Filters and Capacitors, Radio Frequency Interference, General Specification for
MIL-S-15743	14.1	Switches, Rotary, Enclosed
MIL-C-18312	10.1	Capacitor, Fixed, Metallized (Paper, Paper Plastic or Plastic Film) Dielectric, Direct Current (Hermetically Sealed in Metal Cases), General Specification for
MIL-F-18327	21.1	Filter, High Pass, Low Pass, Band Pass, Band Suppression and Dual Functioning, General Specification for

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MIL-R-18546	9.1	Resistor, Fixed, Wirewound (Power Type, Chassis Mounted), General Specification for
MIL-S-19500	6.0	Semiconductor Device, General Specification for
MIL-R-19523	13.1	Relays, Control
MIL-R-19648	13.1	Relay, Time, Delay, Thermal, General Specification for
MIL-C-19978	10.1	Capacitor, Fixed Plastic (or Paper-Plastic) Dielectric (Hermetically Sealed in Metal, Ceramic or Glass Cases), Established and Nonestablished Reliability, General Specification for
MIL-T-21038	11.1	Transformer, Pulse, Low Power, General Specification for
MIL-C-21097	15.1	Connector, Electrical, Printed Wiring Board, General Purpose, General Specification for
MIL-S-21277	14.1	Switches, Liquid Level, General Specification for
MIL-C-21617	15.1	Connectors, Plug and Receptable - Electrical Rectangular, Polarized Shell, Miniature Type
MIL-R-22097	9.1	Resistor, Variable, Nonwirewound (Adjustment Types), General Specification for
MIL-S-22614	14.1	Switches, Sensitive
MIL-R-22684	9.2	Resistor, Fixed, Film, Insulated, General Specification for
MIL-S-22710	14.4	Switches, Code Indicating Wheel (Printed Circuit), (Thumbwheel, In-line and Pushbutton), General Specification for
MIL-S-22885	14.1	Switches, Pushbutton, Illuminated, General Specification for
MIL-C-22992	15.1	Connectors, Plugs and Receptacles, Electrical, Water-Proof, Quick Disconnect, Heavy Duty Type, General Specification for
MIL-C-23183	10.1	Capacitors, Fixed or Variable, Vacuum or Gas Dielectric, General Specification for
MIL-C-23269	10.1	Capacitor, Fixed, Glass Dielectric, Established Reliability, General Specification for
MIL-R-23285	9.1	Resistor, Variable, Nonwirewound, General Specification for
MIL-F-23419	22.1	Fuse, Cartridge, Instrument Type, General Specification for
MIL-T-23648	9.1	Resistor, Thermal, (Thermally Sensitive Resistor), Insulated, General Specification for
MS-24055	15.1	Connector, Plug-Receptacle, Electrical, Hexagonal, 9 Contacts, Female, 7.5 Amps
MS-24056	15.1	Connector, Plug-Receptacle, Electrical, Hexagonal, 9 Contacts, Male, 7.5 Amps

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MIL-C-24308	15.1	Connectors, Electric, Rectangular, Nonenvironmental, Miniature, Polarized Shell, Rack and Panel, General Specification for
MIL-S-24317	14.1	Switches, Multistation, Pushbutton (Illuminated and Non-illuminated), General Specification for
MIL-C-25516	15.1	Connector, Electrical, Miniature, Coaxial, Environment Resistant Type, General Specification for
MIL-C-26482	15.1	Connector, Electrical (Circular, Miniature, Quick Disconnect, Environment Resisting), Receptacles and Plugs, General Specification for
MIL- C-26500	15.1	Connectors, General Purpose, Electrical, Miniature, Circular, Environment Resisting, General Specification for
MIL-R-27208	9.1	Resistor, Variable, Wirewound, Nonprecision, General Specification for
MIL-C-28731	15.1	Connectors, Electrical, Rectangular, Removable Contact, Formed Blade, Fork Type (For Rack and Panel and Other Applications), General Specification for
MIL-C-28748	15.1	Connector, Plug and Receptacle, Rectangular, Rack and Panel, Solder Type and Crimp Type Contacts, General Specification for
MIL-R-28750	13.2	Relay, Solid State, General Specification for
MIL-C-28804	15.1	Connectors, Plug and Receptacle, Electric Rectangular, High Density, Polarized Center Jackscrew, General Specification for, Inactive for New Designs
MIL-C-28840	15.1	Connector, Electrical, Circular Threaded, High Density, High Shock Shipboard, Class D, General Specification for
MIL-M-38510	5.0	Microcircuits, General Specification for
MIL-S-38533	15.3	Sockets, Chip Carrier, Ceramic, General Specification for
MIL-H-38534	5.0	Hybrid Microcircuits, General Specification for
MIL-I-38535	5.0	Integrated Circuits (Microcircuits) Manufacturing, General Specification for
MIL-C-38999	15.1	Connector, Electrical, Circular, Miniature, High Density, Quick Disconnect, (Bayonet, Threaded, and Breech Coupling) Environment Resistant, Removable Crimp and Hermetic Solder Contacts, General Specification for
MIL-C-39001	10.1	Capacitor, Fixed, Mica-Dielectric, Established Reliability, General Specification for
MIL-R-39002	9.1	Resistor, Variable, Wirewound, Semi-Precision, General Specification for
MIL-C-39003	10.1	Capacitor, Fixed, Electrolytic, (Solid Electrolyte), Tantalum, Established Reliability, General Specification for

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MIL-R-39005	9.1	Resistor, Fixed, Wirewound (Accurate), Established Reliability, General Specification for
MIL-C-39006	10.1	Capacitor, Fixed, Electrolytic (Nonsolid Electrolyte) Tantalum Established Reliability, General Specification for
MIL-R-39007	9.1	Resistor, Fixed, Wirewound (Power Type), Established Reliability, General Specification for
MIL-R-39008	9.1	Resistor, Fixed, Composition (Insulated), Established Reliability, General Specification for
MIL-R-39009	9.1	Resistor, Fixed, Wirewound (Power Type, Chassis Mounted) Established Reliability, General Specification for
MIL-C-39010	11.2	Coils, Electrical, Fixed, Radio Frequency, Molded, Established Reliability, General Specification for
MIL-C-39012	15.1	Connector, Coaxial, Radio Frequency, General Specification for
MIL-C-39014	10.1	Capacitor, Fixed, Ceramic Dielectric (General Purpose), Established Reliability, General Specification for
MIL-R-39015	9.1	Resistor, Variable, Wirewound (Lead Screw Actuated), Established Reliability, General Specification for
MIL-R-39016	13.1	Relay, Electromagnetic, Established Reliability, General Specification for
MIL-R-39017	9.1	Resistor, Fixed, Film (Insulated), Established Reliability, General Specification for
MIL-C-39018	10.1	Capacitor, Fixed, Electrolytic (Aluminum Oxide), Established Reliability and Nonestablished Reliability, General Specification for
MIL-C-39019	14.5	Circuit Breakers, Magnetic, Low Power, Sealed, Trip-Free, General Specification for
MIL-C-39022	10.1	Capacitors, Fixed, Metallized, Paper-Plastic Film or Plastic Film Dielectric, Direct and Alternating Current (Hermetically Sealed in Metal or Ceramic Cases), Established Reliability, General Specification for
MIL-R-39023	9.1	Resistor, Variable, Nonwirewound, Precision, General Specification for
MIL-R-39035	9.1	Resistor, Variable, Nonwirewound (Adjustment Type), Established Reliability, General Specification for
MIL-S-45885	14.1	Switch, Rotary
MIL-C-49142	15.1	Connectors, Plugs and Receptacle, Electrical Triaxial, Radio Frequency, General Specification for
MIL-C-55074	15.1	Connectors, Plug and Receptacle, Telephone, Electrical, Subassembly and Accessories and Contact Assembly, Electrical, General Specification for
MIL-P-55110	15.2	Printed Wiring Board, General Specification for
MIL-R-55182	9.1	Resistor, Fixed, Film, Established Reliability, General Specification for

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MIL-C-55235	15.1	Connectors, Coaxial, Radio Frequency, Series TPS
MIL-C-55302	15.1	Connector, Printed Circuit, Subassembly and Accessories
MIL-A-55339	15.1	Adaptors, Connector, Coaxial, Radio Frequency, (Between Series and Within Series), General Specification for
MIL-R-55342	9.1	Resistors, Fixed, Film, Chip, Established Reliability, General Specification for
MIL-C-55365	10.1	Capacitor, Fixed, Electrolytic (Tantalum), Chip, Established Reliability, General Specification for
MIL-S-55433	14.1	Switches, Reed, General Specification for
MIL-C-55514	10.1	Capacitors, Fixed, Plastic (or Metallized Plastic) Dielectric, DC or DC-AC, In Non-Metal Cases, Established Reliability, General Specification for
MIL-C-55629	14.5	Circuit Breaker, Magnetic, Unsealed, or Panel Seal, Trip-Free, General Specification for
MIL-T-55631	11.1	Transformer, Intermediate Frequency, Radio Frequency and Discriminator, General Specification for
MIL-C-55681	10.1	Capacitor, Chip, Multiple Layer, Fixed, Unencapsulated Ceramic Dielectric, Established Reliability, General Specification for
MIL-C-81511	15.1	Connector, Electrical, Circular, High Density, Quick Disconnect, Environment Resisting and Accessories, General Specification for
MIL-S-81551	14.1	Switches; Toggle, Hermetically Sealed, General Specification for
MIL-C-81659	15.1	Connectors, Electrical Rectangular, Crimp Contact
MIL-S-82359	14.1	Switch, Rotary, Variable Resistor Assembly Type
MIL-C-83383	14.5	Circuit Breaker, Remote Control, Thermal, Trip-Free, General Specification for
MIL-R-83401	9.1	Resistor Networks, Fixed, Film and Capacitor-Resistor Networks, Ceramic Capacitors and Fixed Film Resistors, General Specification for
MIL-C-83421	10.1	Capacitors, Fixed Metallized Plastic Film Dielectric (DC, AC or DC and AC) Hermetically Sealed in Metal or Ceramic Cases, Established Reliability, General Specification for
MIL-C-83446	11.2	Coils, Radio Frequency, Chip, Fixed or Variable, General Specification for
MIL-C-83500	10.1	Capacitor, Fixed, Electrolytic (Nonsolid Electrolyte), Tantalum Cathode, General Specification for
MIL-S-83504	14.1	Switches, Dual In-Line Package (DIP), General Specification for
MIL-C-83513	15.1	Connector, Electrical, Rectangular, Microminiature, Polarized Shell, General Specification for

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MIL-C-83515	15.1	Connectors, Telecommunication, Polarized Shell, General Specification for
MIL-R-83516	13.1	Relays, Reed, Dry, General Specification for
MIL-C-83517	15.1	Connectors, Coaxial, Radio Frequency for Coaxial, Strip or Microstrip Transmission Line, General Specification for
MIL-R-83520	13.1	Relays, Electromechanical, General Purpose, Non-Hermetically Sealed, Plastic Enclosure (Dust Cover), General Specification for
MIL-C-83527	15.1	Connectors, Plug and Receptacle, Electrical, Rectangular Multiple Insert Type, Rack to Panel, Environment Resisting, 150°C Total Continuous Operating Temperature, General Specification for
MIL-R-83536	13.1	Relays, Electromagnetic, Established Reliability, General Specification for
MIL-C-83723	15.1	Connector, Electrical (Circular Environment Resisting), Receptacles and Plugs, General Specification for
MIL-R-83725	13.1	Relay, Vacuum, General Specification for
MIL-R-83726	13.1, 13.2, 13.3	Relays, Hybrid and Solid State, Time Delay, General Specification for
MIL-S-83731	14.1	Switch, Toggle, Unsealed and Sealed Toggle, General Specification for
MIL-C-83733	15.1	Connector, Electrical, Miniature, Rectangular Type, Rack to Panel, Environment Resisting, 200°C Total Continuous Operating Temperature, General Specification for
MIL-S-83734	15.3	Sockets, Plug-In Electronic Components, Dual-In-Line (DIPS) and Single-In-Line Packages (SIPS), General Specification for
MIL-C-85028	15.1	Connector, Electrical, Rectangular, Individual Contact Sealing, Polarized Center Jackscrew, General Specification for

STANDARD	TITLE
MIL-STD-756	Reliability Modeling and Prediction
MIL-STD-883	Test Methods and Procedures for Microelectronics
MIL-STD-975	NASA Standard Electrical, Electronic and Electromechanical (EEE) Parts List
MIL-STD-1547	Electronic Parts, Materials and Processes for Space and Launch Vehicles, Technical Requirements for
MIL-STD-1772	Certification Requirements for Hybrid Microcircuit Facilities and Lines

Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer. Single copies are also available (without charge) upon written request to:

Standardization Document Order Desk, 700 Robins Ave., Building 4, Section D, Philadelphia, PA 19111-5094, (215) 697-2667

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5.1 MICROCIRCUITS, GATE/LOGIC ARRAYS AND MICROPROCESSORS

DESCRIPTION

1. Bipolar Devices, Digital and Linear Gate/Logic Arrays
2. MOS Devices, Digital and Linear Gate/Logic Arrays
3. Field Programmable Logic Array (PLA) and Programmable Array Logic (PAL)
4. Microprocessors

$$\lambda_p = (C_1\pi_T + C_2\pi_E)\,\pi_Q\pi_L \quad \text{Failures}/10^6 \text{ Hours}$$

Bipolar Digital and Linear Gate/Logic Array Die Complexity Failure Rate - C_1

Digital		Linear		PLA/PAL	
No. Gates	C_1	No. Transistors	C_1	No. Gates	C_1
1 to 100	.0025	1 to 100	.010	Up to 200	.010
101 to 1,000	.0050	101 to 300	.020	201 to 1,000	.021
1,001 to 3,000	.010	301 to 1,000	.040	1,001 to 5,000	.042
3,001 to 10,000	.020	1,001 to 10,000	.060		
10,001 to 30,000	.040				
30,001 to 60,000	.080				

MOS Linear and Digital Gate/Logic Array Die Complexity Failure Rate - C_1*

Digital		Linear		PLA/PAL	
No. Gates	C_1	No. Transistors	C_1	No. Gates	C_1
1 to 100	.010	1 to 100	.010	Up to 500	.00085
101 to 1,000	.020	101 to 300	.020	501 to 1,000	.0017
1,001 to 3,000	.040	301 to 1,000	.040	2,001 to 5,000	.0034
3,001 to 10,000	.080	1,001 to 10,000	.060	5,001 to 20,000	.0068
10,001 to 30,000	.16				
30,001 to 60,000	.29				

*NOTE: For CMOS gate counts above 60,000 use the VHSIC/VHSIC-Like model in Section 5.3

Microprocessor
Die Complexity Failure Rate - C_1

No. Bits	Bipolar C_1	MOS C_1
Up to 8	.060	.14
Up to 16	.12	.28
Up to 32	.24	.56

All Other Model Parameters

Parameter	Refer to
π_T	Section 5.8
C_2	Section 5.9
π_E, π_Q, π_L	Section 5.10

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5.2 MICROCIRCUITS, MEMORIES

DESCRIPTION

1. Read Only Memories (ROM)
2. Programmable Read Only Memories (PROM)
3. Ultraviolet Eraseable PROMs (UVEPROM)
4. "Flash," MNOS and Floating Gate Electrically Eraseable PROMs (EEPROM). Includes both floating gate tunnel oxide (FLOTOX) and textured polysilicon type EEPROMs
5. Static Random Access Memories (SRAM)
6. Dynamic Random Access Memories (DRAM)

$$\lambda_p = (C_1 \pi_T + C_2 \pi_E + \lambda_{cyc}) \pi_Q \pi_L \quad \text{Failures}/10^6 \text{ Hours}$$

Die Complexity Failure Rate - C_1

Memory Size, B (Bits)	MOS				Bipolar	
	ROM	PROM, UVEPROM, EEPROM, EAPROM	DRAM	SRAM (MOS & BiMOS)	ROM, PROM	SRAM
Up to 16K	.00065	.00085	.0013	.0078	.0094	.0052
16K < B ≤ 64K	.0013	.0017	.0025	.016	.019	.011
64K < B ≤ 256K	.0026	.0034	.0050	.031	.038	.021
256K < B ≤ 1M	.0052	.0068	.010	.062	.075	.042

A_1 Factor for λ_{cyc} Calculation

Total No. of Programming Cycles Over EEPROM Life, C	Flotox[1]	Textured-Poly[2]
Up to 100	.00070	.0097
100 < C ≤ 200	.0014	.014
200 < C ≤ 500	.0034	.023
500 < C ≤ 1K	.0068	.033
1K < C ≤ 3K	.020	.061
3K < C ≤ 7K	.049	.14
7K < C ≤ 15K	.10	.30
15K < C ≤ 20K	.14	.30
20K < C ≤ 30K	.20	.30
30K < C ≤ 100K	.68	.30
100K < C ≤ 200K	1.3	.30
200K < C ≤ 400K	2.7	.30
400K < C ≤ 500K	3.4	.30

1. $A_1 = 6.817 \times 10^{-6}$ (C)
2. No underlying equation for Textured-Poly.

A_2 Factor for λ_{cyc} Calculation

Total No. of Programming Cycles Over EEPROM Life, C	Textured-Poly A_2
Up to 300K	0
300K < C ≤ 400K	1.1
400K < C ≤ 500K	2.3

All Other Model Parameters

Parameter	Refer to
π_T	Section 5.8
C_2	Section 5.9
π_E, π_Q, π_L	Section 5.10
λ_{cyc} (EEPROMS only)	Page 5-5

$\lambda_{cyc} = 0$ For all other devices

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EEPROM Read/Write Cycling Induced Failure Rate - λ_{cyc}

All Memory Devices Except Flotox and Textured-Poly EEPROMs	$\lambda_{cyc} = 0$
Flotox and Textured Poly EEPROMs	$\lambda_{cyc} = \left[A_1 B_1 + \frac{A_2 B_2}{\pi_Q} \right] \pi_{ECC}$

Model Factor	Flotox	Textured-Poly
A_1	Page 5-4	Page 5-4
B_1	Page 5-6	Page 5-6
A_2	$A_2 = 0$	Page 5-5
B_2	$B_2 = 0$	Page 5-6
π_Q	Section 5.10	Section 5.10
Error Correction Code (ECC) Options:		
1. No On-Chip ECC	$\pi_{ECC} = 1.0$	$\pi_{ECC} = 1.0$
2. On-Chip Hamming Code	$\pi_{ECC} = .72$	$\pi_{ECC} = .72$
3. Two-Needs-One Redundant Cell Approach	$\pi_{ECC} = .68$	$\pi_{ECC} = .68$

NOTES:

1. See Reference 24 for modeling off-chip error detection and correction schemes at the memory system level.
2. If EEPROM type is unknown, assume Flotox.
3. Error Correction Code Options: Some EEPROM manufacturers have incorporated on-chip error correction circuitry into their EEPROM devices. This is represented by the on-chip hamming code entry. Other manufacturers have taken a redundant cell approach which incorporates an extra storage transistor in every memory cell. This is represented by the two-needs-one redundant cell entry.
4. The A_1 and A_2 factors shown in Section 5.2 were developed based on an assumed system life of 10,000 operating hours. For EEPROMs used in systems with significantly longer or shorter expected lifetimes the A_1 and A_2 factors should be multiplied by:

$$\frac{\text{System Lifetime Operating Hours}}{10{,}000}$$

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5.2 MICROCIRCUITS, MEMORIES

B_1 and B_2 Factors for λ_{cyc} Calculation

Memory Size, B(Bits) →	Flotox[1] (B_1)				Textured-Poly[2] (B_1)				Textured-Poly[3] (B_2)			
T_J (°C)	4K	16K	64K	256K	1M	4K	16K	64K	256K	1M	4K	16K	64K	256K	1M
25	.27	0.55	1.1	2.2	4.3	.47	.66	.94	1.3	1.9	.54	0.76	1.1	1.5	2.1
30	.30	0.60	1.2	2.4	4.8	.50	.71	1.0	1.4	2.0	.50	0.71	1.0	1.4	2.0
35	.33	0.66	1.3	2.7	5.2	.54	.77	1.1	1.5	2.2	.47	0.67	.95	1.3	1.9
40	.36	0.72	1.4	2.9	5.7	.58	.82	1.2	1.6	2.3	.45	0.63	.89	1.3	1.8
45	.40	0.79	1.6	3.2	6.3	.62	.88	1.3	1.8	2.5	.42	0.59	.84	1.2	1.7
50	.43	0.86	1.7	3.4	6.8	.67	.95	1.3	1.9	2.7	.40	0.56	.80	1.1	1.6
55	.47	0.93	1.9	3.7	7.4	.71	1.0	1.4	2.0	2.8	.38	0.53	.75	1.1	1.5
60	.51	1.0	2.0	4.1	8.0	.76	1.1	1.5	2.1	3.0	.36	0.50	.72	1.0	1.4
65	.55	1.1	2.2	4.4	8.6	.81	1.1	1.6	2.3	3.2	.34	0.48	.68	.96	1.3
70	.59	1.2	2.4	4.7	9.3	.86	1.2	1.7	2.4	3.4	.32	0.45	.65	.91	1.3
75	.63	1.3	2.5	5.1	10	.91	1.3	1.8	2.6	3.6	.31	0.43	.62	.87	1.2
80	.68	1.4	2.7	5.4	11	.96	1.4	1.9	2.7	3.8	.29	0.41	.59	.83	1.2
85	.73	1.5	2.9	5.8	12	1.0	1.4	2.0	2.9	4.0	.28	0.39	.56	.79	1.1
90	.78	1.6	3.1	6.2	12	1.1	1.5	2.2	3.0	4.3	.27	0.38	.54	.75	1.1
95	.83	1.7	3.3	6.7	13	1.1	1.6	2.3	3.2	4.5	.26	0.36	.51	.72	1.0
100	.89	1.8	3.5	7.1	14	1.2	1.7	2.4	3.4	4.7	.25	0.35	.49	.69	.98
105	.94	1.9	3.8	7.5	15	1.3	1.8	2.5	3.5	5.0	.24	0.33	.47	.66	.94
110	1.0	2.0	4.0	8.0	16	1.3	1.9	2.6	3.7	5.2	.23	0.32	.45	.64	.90
115	1.1	2.1	4.2	8.5	17	1.4	1.9	2.8	3.9	5.5	.22	0.31	.44	.61	.86
120	1.1	2.2	4.5	9.0	18	1.4	2.0	2.9	4.1	5.7	.21	0.30	.42	.59	.83
125	1.2	2.4	4.7	9.5	19	1.5	2.1	3.0	4.3	6.0	.20	0.29	.41	.57	.80
130	1.3	2.5	5.0	10	20	1.6	2.2	3.2	4.4	6.3	.19	0.27	.39	.55	.77
135	1.3	2.6	5.3	11	21	1.6	2.3	3.3	4.6	6.5	.19	0.27	.38	.53	.75
140	1.4	2.8	5.6	11	22	1.7	2.4	3.4	4.8	6.8	.18	0.26	.36	.51	.72
145	1.5	2.9	5.8	12	23	1.8	2.5	3.6	5.0	7.1	.18	0.25	.35	.50	.70
150	1.5	3.1	6.1	12	24	1.9	2.6	3.7	5.2	7.4	.17	0.24	.34	.48	.68
155	1.6	3.2	6.4	13	26	1.9	2.7	3.9	5.4	7.7	.16	0.23	.33	.46	.65
160	1.7	3.4	6.8	14	27	2.0	2.8	4.0	5.6	8.0	.16	0.23	.32	.45	.63
165	1.8	3.5	7.1	14	28	2.1	2.9	4.2	5.9	8.2	.15	0.22	.31	.44	.61
170	1.9	3.7	7.4	15	29	2.2	3.0	4.3	6.1	8.6	.15	0.21	.30	.42	.60
175	1.9	3.9	7.7	15	31	2.2	3.1	4.5	6.3	8.9	.15	0.21	.29	.41	.58

1. $B_1 = \left(\frac{B}{16000}\right)^{.5}\left[\exp\left(\frac{-.15}{8.617 \times 10^{-5}}\left(\frac{1}{T_J + 273} - \frac{1}{333}\right)\right)\right]$

2. $B_1 = \left(\frac{B}{64000}\right)^{.25}\left[\exp\left(\frac{-.12}{8.617 \times 10^{-5}}\left(\frac{1}{T_J + 273} - \frac{1}{303}\right)\right)\right]$

3. $B_2 = \left(\frac{B}{64000}\right)^{.25}\left[\exp\left(\frac{-.1}{8.617 \times 10^{-5}}\left(\frac{1}{T_J + 273} - \frac{1}{303}\right)\right)\right]$

T_J = Worse Case Junction Temperature (°C). See Section 5.11 for T_J Determination

B = Number of bits. NOTE: 1K = 1024 bits

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5.5 MICROCIRCUITS, HYBRIDS

DESCRIPTION
Hybrid Microcircuits

$$\lambda_p = [\Sigma N_C \lambda_C] (1 + .2 \pi_E) \pi_F \pi_Q \pi_L \text{ Failures/}10^6 \text{ Hours}$$

N_C = Number of Each Particular Component
λ_C = Failure Rate of Each Particular Component

The general procedure for developing an overall hybrid failure rate is to calculate an individual failure rate for each component type used in the hybrid and then sum them. This summation is then modified to account for the overall hybrid function (π_F), screening level (π_Q), and maturity (π_L). The hybrid package failure rate is a function of the active component failure modified by the environmental factor (i.e., (1 + .2 π_E)). Only the component types listed in the following table are considered to contribute significantly to the overall failure rate of most hybrids. All other component types (e.g., resistors, inductors, etc.) are considered to contribute insignificantly to the overall hybrid failure rate, and are assumed to have a failure rate of zero. This simplification is valid for most hybrids; however, if the hybrid consists of mostly passive components then a failure rate should be calculated for these devices. If factoring in other component types, assume π_Q = 1, π_E =1 and T_A = Hybrid Case Temperature for these calculations.

Determination of λ_C

Determine λ_C for These Component Types	Handbook Section	Make These Assumptions When Determining λ_C
Microcircuits	5	C_2 = 0, π_Q = 1, π_L = 1, T_J as Determined from Section 5.12, λ_{BP} = 0 (for VHSIC).
Discrete Semiconductors	6	π_Q = 1, π_A = 1, T_J as Determined from Section 6.14, π_E = 1.
Capacitors	10	π_Q = 1, T_A = Hybrid Case Temperature, π_E = 1.
NOTE: If maximum rated stress for a die is unknown, assume the same as for a discretely package die of the same type. If the same die has several ratings based on the discrete packaged type, assume the lowest rating. Power rating used should be based on case temperature for discrete semiconductors.		

Circuit Function Factor - π_F

Circuit Type	π_F
Digital	1.0
Video, 10 MHz < f < 1 GHz	1.2
Microwave, f > 1 GHz	2.6
Linear, f < 10 MHz	5.8
Power	21

All Other Hybrid Model Parameters

π_L, π_Q, π_E	Refer to Section 5.10

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ines electrolytic: deoxidized percentages of other elements. Unalloyed copper (min

5.6 MICROCIRCUITS, SAW DEVICES

DESCRIPTION
Surface Acoustic Wave Devices

$$\lambda_p = 2.1\ \pi_Q\ \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Quality Factor - π_Q

Screening Level	π_Q
10 Temperature Cycles (-55°C to +125°C) with end point electrical tests at temperature extremes.	.10
None beyond best commerical practices.	1.0

Environmental Factor - π_E

Environment	π_E
G_B	.5
G_F	2.0
G_M	4.0
N_S	4.0
N_U	6.0
A_{IC}	4.0
A_{IF}	5.0
A_{UC}	5.0
A_{UF}	8.0
A_{RW}	8.0
S_F	.50
M_F	5.0
M_L	12
C_L	220

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5.13 MICROCIRCUITS, EXAMPLES

	LM106	LM741A	Si NPN	Si PNP	Si Diode	Source
No. of Pins	8	14	3	3	2	Vendor Spec. Sheet
Power Dissipation, P_D (W)	.33	.35	.6	.6	.42	Circuit Analysis
Area of Chip (in.^{2})	.0041	.0065	.0025	.0025	.0022	Equ. 2 Above
θ_{JC} (°C/W)	30.8	19.4	50.3	50.3	56.3	Equ. 1 Above
T_J (°C)	75	72	95	95	89	Equ. 3 Above

2. Calculate Failure Rates for Each Component:

A) LM106 Die, 13 Transistors (from Vendor Spec. Sheet)

$\lambda_p = [C_1 \pi_T + C_2 \pi_E] \pi_Q \pi_L$ Section 5.1

Because $C_2 = 0$;

$\lambda_p = C_1 \pi_T \pi_Q \pi_L$ π_T: Section 5.8; π_Q, π_L Default to 1.0

$= (.01)(3.8)(1)(1) = .038$ Failures/10^6 Hours

B) LM741 Die, 23 Transistors. Use Same Procedure as Above.

$\lambda_p = C_1 \pi_T \pi_Q \pi_L = (.01)(3.1)(1)(1) = .031$ Failures/10^6 Hours

C) Silicon NPN Transistor, Rated Power = 5W (From Vendor Spec. Sheet), $V_{CE}/V_{CEO} = .6$, Linear Application

$\lambda_p = \lambda_b \pi_T \pi_A \pi_R \pi_S \pi_Q \pi_E$ Section 6.3; π_A, π_Q, π_E Default to 1.0

$= (.00074)(3.9)(1.0)(1.8)(.29)(1)(1)$

$= .0015$ Failures/10^6 Hours

D) Silicon PNP Transistor, Same as C.

$\lambda_p = .0015$ Failures/10^6 Hours

E) Silicon General Purpose Diode (Analog), Voltage Stress = 60%, Metallurgically Bonded Construction.

$\lambda_p = \lambda_b \pi_T \pi_S \pi_C \pi_Q \pi_E$ Section 6.1; π_Q, π_E Default to 1.0

$= (.0038)(6.3)(.29)(1)(1)(1)$

$= .0069$ Failures/10^6 Hours

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5.13 MICROCIRCUITS, EXAMPLES

F) Ceramic Chip Capacitor, Voltage Stress = 50%,
$T_A = T_{CASE}$ for the Hybrid, 1340 pF, 125°C Rated Temp.

$\lambda_p = \lambda_b \pi_{CV} \pi_Q \pi_E$ Section 10.11; π_Q, π_E Default to 1.0
$= (.0028)(1.4)(1)(1)$
$= .0039$ Failures/10^6 Hours

G) Thick Film Resistors, per instructions in Section 5.5, the contribution of these devices is considered insignificant relative to the overall hybrid failure rate and they may be ignored.

Overall Hybrid Part Failure Rate Calculation:

$\lambda_p = [\Sigma N_C \lambda_c](1 + .2 \pi_E) \pi_F \pi_Q \pi_L$

π_E	= 6.0	Section 5.10
π_F	= 5.8	Section 5.5
π_Q	= 1	Section 5.10
π_L	= 1	Section 5.10

$\lambda_p = [(1)(.038) + (1)(.031) + (2)(.0015) + (2)(.0015)$
$+ (2)(.0069) + (2)(.0039)](1 + .2(6.0))(5.8)(1)(1)$

$\lambda_p = 1.2$ Failures/10^6 Hours

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6.0 DISCRETE SEMICONDUCTORS, INTRODUCTION

The semiconductor transistor, diode and opto-electronic device sections present the failure rates on the basis of device type and construction. An analytical model of the failure rate is also presented for each device category. The various types of discrete semiconductor devices require different failure rate models that vary to some degree. The models apply to single devices unless otherwise noted. For multiple devices in a single package the hybrid model in Section 5.5 should be used.

The applicable MIL specification for transistors, and optoelectronic devices is MIL-S-19500. The quality levels (JAN, JANTX, JANTXV) are as defined in MIL-S-19500.

The temperature factor (π_T) is based on the device junction temperature. Junction temperature should be computed based on worse case power (or maximum power dissipation) and the device junction to case thermal resistance. Determination of junction temperatures is explained in Section 6.14.

Reference 28 should be consulted for further detailed information on the models appearing in this section.

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6.1 DIODES, LOW FREQUENCY

SPECIFICATION
MIL-S-19500

DESCRIPTION
Low Frequency Diodes: General Purpose Analog, Switching, Fast Recovery, Power Rectifier, Transient Suppressor, Current Regulator, Voltage Regulator, Voltage Reference

$$\lambda_p = \lambda_b \pi_T \pi_S \pi_C \pi_Q \pi_E \quad \text{Failures/}10^6 \text{ Hours}$$

Base Failure Rate - λ_b

Diode Type/Application	λ_b
General Purpose Analog	.0038
Switching	.0010
Fast Recovery Power Rectifier	.025
Power Rectifier/Schottky	.0030
Power Diode	
Power Rectifier with High Voltage Stacks	.0050/ Junction
Transient Suppressor/Varistor	.0013
Current Regulator	.0034
Voltage Regulator and Voltage Reference (Avalanche and Zener)	.0020

Temperature Factor - π_T
(General Purpose Analog, Switching, Fast Recovery, Power Rectifier, Transient Suppressor)

T_J (°C)	π_T	T_J (°C)	π_T
25	1.0	105	9.0
30	1.2	110	10
35	1.4	115	11
40	1.6	120	12
45	1.9	125	14
50	2.2	130	15
55	2.6	135	16
60	3.0	140	18
65	3.4	145	20
70	3.9	150	21
75	4.4	155	23
80	5.0	160	25
85	5.7	165	28
90	6.4	170	30
95	7.2	175	32
100	8.0		

$$\pi_T = \exp\left(-3091\left(\frac{1}{T_J + 273} - \frac{1}{298}\right)\right)$$

T_J = Junction Temperature (°C)

Temperature Factor - π_T
(Voltage Regulator, Voltage Reference, and Current Regulator)

T_J (°C)	π_T	T_J (°C)	π_T
25	1.0	105	3.9
30	1.1	110	4.2
35	1.2	115	4.5
40	1.4	120	4.8
45	1.5	125	5.1
50	1.6	130	5.4
55	1.8	135	5.7
60	2.0	140	6.0
65	2.1	145	6.4
70	2.3	150	6.7
75	2.5	155	7.1
80	2.7	160	7.5
85	3.0	165	7.9
90	3.2	170	8.3
95	3.4	175	8.7
100	3.7		

$$\pi_T = \exp\left(-1925\left(\frac{1}{T_J + 273} - \frac{1}{298}\right)\right)$$

T_J = Junction Temperature (°C)

Supersedes page 6-2 of Revision F
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DESCRIPTION
Traveling Wave Tubes

$$\lambda_p = \lambda_b \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - λ_b

Power (W)	Frequency (GHz) .1	1	2	4	6	8	10	14	18
10	11	12	13	16	19	24	29	42	61
100	11	12	13	16	20	24	29	42	61
500	11	12	13	16	20	24	29	42	61
1000	11	12	13	16	20	24	29	42	62
3000	11	12	14	17	20	24	29	43	63
5000	12	13	14	17	20	25	30	44	64
8000	12	13	14	17	21	26	31	45	66
10000	12	13	15	18	22	26	32	46	68
15000	13	14	15	19	23	27	33	49	71
20000	14	15	16	20	24	29	35	51	75
30000	15	16	18	22	26	32	39	56	83
40000	17	18	20	24	29	35	43	62	91

$\lambda_b = 11(1.00001)^P (1.1)^F$

P = Rated Power in Watts (Peak, if Pulsed), $.001 \leq P \leq 40{,}000$

F = Operating Frequency in GHz, $.1 \leq F \leq 18$

If the operating frequency is a band, or two different values, use the geometric mean of the end point frequencies when using table.

Environment Factor - π_E

Environment	π_E
G_B	.5
G_F	1.5
G_M	7.0
N_S	3.0
N_U	10
A_{IC}	5.0
A_{IF}	7.0
A_{UC}	6.0
A_{UF}	9.0
A_{RW}	20
S_F	.05
M_F	11
M_L	33
C_L	500

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7.3 TUBES, MAGNETRON

DESCRIPTION
Magnetrons, Pulsed and Continuous Wave (CW)

$$\lambda_p = \lambda_b \pi_U \pi_C \pi_E \quad \text{Failures/}10^6 \text{ Hours}$$

Base Failure Rate - λ_b

P(MW)	Frequency (GHz) .1	.5	1	5	10	20	30	40	50	60	70	80	90	100
.01	1.4	4.6	7.6	24	41	67	91	110	130	150	170	190	200	220
.05	1.9	6.3	10	34	56	93	120	150	180	210	230	260	280	300
.1	2.2	7.2	12	39	64	110	140	180	210	240	270	290	320	350
.3	2.8	9.0	15	48	80	130	180	220	260	300	330	370	400	430
.5	3.1	10	17	54	89	150	200	240	290	330	370	410	440	480
1	3.5	11	19	62	100	170	230	280	330	380	420	470	510	550
3	4.4	14	24	77	130	210	280	350	410	470	530	580	630	680
5	4.9	16	26	85	140	230	310	390	460	520	580	640	700	760

Pulsed Magnetrons:

$\lambda_b = 19(F)^{.73}(P)^{.20}$

F = Operating Frequency in GHz, $.1 \le F \le 100$

P = Output Power in MW, $.01 \le P \le 5$

CW Magnetrons (Rated Power < 5 KW):

$\lambda_b = 18$

Utillization Factor - π_U

Utilization (Radiate Hours/ Filament Hours)	π_U
0.0	.44
0.1	.50
0.2	.55
0.3	.61
0.4	.66
0.5	.72
0.6	.78
0.7	.83
0.8	.89
0.9	.94
1.0	1.0

$\pi_U = 0.44 + 0.56R$

R = Radiate Hours/Filament Hours

Construction Factor - π_C

Construction	π_C
CW (Rated Power < 5 KW)	1.0
Coaxial Pulsed	1.0
Conventional Pulsed	5.4

Environment Factor - π_E

Environment	π_E
G_B	1.0
G_F	2.0
G_M	4.0
N_S	15
N_U	47
A_{IC}	10
A_{IF}	16
A_{UC}	12
A_{UF}	23
A_{RW}	80
S_F	.50
M_F	43
M_L	133
C_L	2000

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$$\lambda_p = \lambda_b \pi_T \pi_P \pi_S \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Resistor Style	Specification MIL-R-	Description	λ_b	π_T Table Use Column:	π_S Table Use Column:
RC	11	Resistor, Fixed, Composition (Insulated)	.0017	1	2
RCR	39008	Resistor, Fixed, Composition (Insulated) Est. Rel.	.0017	1	2
RL	22684	Resistor, Fixed, Film, Insulated	.0037	2	1
RLR	39017	Resistor, Fixed, Film (Insulated), Est. Rel.	.0037	2	1
RN (R, C or N)	55182	Resistor, Fixed, Film, Established Reliability	.0037	2	1
RM	55342	Resistor, Fixed, Film, Chip, Established Reliability	.0037	2	1
RN	10509	Resistor, Fixed Film (High Stability)	.0037	2	1
RD	11804	Resistor, Fixed, Film (Power Type)	.0037	N/A, $\pi_T = 1$	1
RZ	83401	Resistor Networks, Fixed, Film	.0019	1	N/A, $\pi_S = 1$
RB	93	Resistor, Fixed, Wirewound (Accurate)	.0024	2	1
RBR	39005	Resistor, Fixed, Wirewound (Accurate) Est. Rel.	.0024	2	1
RW	26	Resistor, Fixed, Wirewound (Power Type)	.0024	2	2
RWR	39007	Resistor, Fixed, Wirewound (Power Type) Est. Rel.	.0024	2	2
RE	18546	Resistor, Fixed, Wirewound (Power Type, Chassis Mounted)	.0024	2	2
RER	39009	Resistor, Fixed, Wirewound (Power Type, Chassis Mounted) Est. Rel.	.0024	2	2
RTH	23648	Thermistor, (Thermally Sensitive Resistor), Insulated	.0019	N/A, $\pi_T = 1$	N/A, $\pi_S = 1$
RT	27208	Resistor, Variable, Wirewound (Lead Screw Activated)	.0024	2	1
RTR	39015	Resistor, Variable, Wirewound (Lead Screw Activated), Established Reliability	.0024	2	1
RR	12934	Resistor, Variable, Wirewound, Precision	.0024	2	1
RA	19	Resistor, Variable, Wirewound (Low Operating Temperature)	.0024	1	1
RK	39002	Resistor, Variable, Wirewound, Semi-Precision	.0024	1	1
RP	22	Resistor, Wirewound, Power Type	.0024	2	1
RJ	22097	Resistor, Variable, Nonwirewound	.0037	2	1
RJR	39035	Resistor, Variable, Nonwirewound Est. Rel.	.0037	2	1
RV	94	Resistor, Variable, Composition	.0037	2	1
RQ	39023	Resistor, Variable, Nonwirewound, Precision	.0037	1	1
RVC	23285	Resistor, Variable, Nonwirewound	.0037	1	1

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9.1 RESISTORS

Temperature Factor - π_T

T(°C)	Column 1	Column 2
20	.88	.95
30	1.1	1.1
40	1.5	1.2
50	1.8	1.3
60	2.3	1.4
70	2.8	1.5
80	3.4	1.6
90	4.0	1.7
100	4.8	1.9
110	5.6	2.0
120	6.6	2.1
130	7.6	2.3
140	8.7	2.4
150	10	2.5

$$\pi_T = \exp\left(\frac{-Ea}{8.617 \times 10^{-5}}\left(\frac{1}{T + 273} - \frac{1}{298}\right)\right)$$

Column 1: Ea = .2

Column 2: Ea = .08

T = Resistor Case Temperature. Can be approximated as ambient component temperature for low power dissipation non-power type resistors.

NOTE: π_T values shown should only be used up to the temperature rating of the device. For devices with ratings higher than 150°C, use the equation to determine π_T.

Power Factor - π_P

Power Dissipation (Watts)	π_P
.001	.068
.01	.17
.13	.44
.25	.58
.50	.76
.75	.89
1.0	1.0
2.0	1.3
3.0	1.5
4.0	1.7
5.0	1.9
10	2.5
25	3.5
50	4.6
100	6.0
150	7.1

$$\pi_P = (\text{Power Dissipation})^{.39}$$

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9.1 RESISTORS

Power Stress Factor - π_S

Power Stress	Column 1	Column 2
.1	.79	.66
.2	.88	.81
.3	.99	1.0
.4	1.1	1.2
.5	1.2	1.5
.6	1.4	1.8
.7	1.5	2.3
.8	1.7	2.8
.9	1.9	3.4

Column 1: $\pi_S = .71e^{1.1(S)}$

Column 2: $\pi_S = .54e^{2.04(S)}$

$$S = \frac{\text{Actual Power Dissipation}}{\text{Rated Power}}$$

Environment Factor - π_E

Environment	π_E
G_B	1.0
G_F	4.0
G_M	16
N_S	12
N_U	42
A_{IC}	18
A_{IF}	23
A_{UC}	31
A_{UF}	43
A_{RW}	63
S_F	.50
M_F	37
M_L	87
C_L	1728

Quality Factor - π_Q

Quality	π_Q
Established Reliability Styles	
S	.03
R	0.1
P	0.3
M	1.0
Non-Established Reliability Resistors (Most Two-Letter Styles)	3.0
Commercial or Unknown Screening Level	10

NOTE: Established reliability styles are failure rate graded (S, R, P, M) based on life testing defined in the applicable military device specification. This category usually applies only to three-letter styles with an "R" suffix.

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10.1 CAPACITORS

$$\lambda_p = \lambda_b \pi_T \pi_C \pi_V \pi_{SR} \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Capacitor Style	Spec. MIL-C-	Description	λ_b	π_T Table - Use Column:	π_C Table - Use Column:	π_V Table - Use Column:	π_{SR}
CP	25	Capacitor, Fixed, Paper-Dielectric, Direct Current (Hermetically Sealed in Metal Cases)	.00037	1	1	1	1
CA	12889	Capacitor, By-Pass, Radio - Interference Reduction, Paper Dielectric, AC and DC (Hermetically sealed in Metallic Cases)	.00037	1	1	1	1
CZ, CZR	11693	Capacitor, Feed through, Radio Interference Reduction AC and DC (Hermetically sealed in metal cases), Established and Nonestablished Reliability	.00037	1	1	1	1
CQ, CQR	19978	Capacitor, Fixed Plastic (or Paper-Plastic) Dielectric (Hermetically sealed in metal, ceramic or glass cases), Established and Nonestablished Reliability	.00051	1	1	1	1
CH	18312	Capacitor, Fixed, Metallized (Paper, Paper Plastic or Plastic Film) Dielectric, Direct Current (Hermetically Sealed in Metal Cases)	.00037	1	1	1	1
CHR	39022	Capacitor, Fixed, Metallized Paper, Paper-Plastic Film or Plastic Film Dielectric	.00051	1	1	1	1
CFR	55514	Capacitor, Fixed, Plastic (or Metallized Plastic) Dielectric, Direct Current in Non-Metal Cases	.00051	1	1	1	1
CRH	83421	Capacitor, Fixed Supermetallized Plastic Film Dielectric (DC, AC or DC and AC) Hermetically Sealed in Metal Cases, Established Reliability	.00051	1	1	1	1
CM	5	Capacitors, Fixed, Mica Dielectric	.00076	2	1	2	1
CMR	39001	Capacitor, Fixed, Mica Dielectric, Established Reliability	.00076	2	1	2	1
CB	10950	Capacitor, Fixed, Mica Dielectric, Button Style	.00076	2	1	2	1
CY	11272	Capacitor, Fixed, Glass Dielectric	.00076	2	1	2	1
CYR	23269	Capacitor, Fixed, Glass Dielectric, Established Reliability	.00076	2	1	2	1

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10.1 CAPACITORS

Capacitor Style	Spec. MIL-C-	Description	λ_b	π_T Table - Use Column:	π_C Table - Use Column:	π_V Table - Use Column:	π_{SR}
CK	11015	Capacitor, Fixed, Ceramic Dielectric (General Purpose)	.00099	2	1	3	1
CKR	39014	Capacitor, Fixed, Ceramic Dielectric (General Purpose), Established Reliability	.00099	2	1	3	1
CC, CCR	20	Capacitor, Fixed, Ceramic Dielectric (Temperature Compensating), Established and Nonestablished Reliability	.00099	2	1	3	1
CDR	55681	Capacitor, Chip, Multiple Layer, Fixed, Ceramic Dielectric, Established Reliability	.0020	2	1	3	1
CSR	39003	Capacitor, Fixed, Electrolytic (Solid Electrolyte), Tantalum, Established Reliability	.00040	1	2	4	See π_{SR} Table
CWR	55365	Capacitor, Fixed, Electrolytic (Tantalum), Chip, Established Reliability	.00005	1	2	4	See π_{SR} Table
CL	3965	Capacitor, Fixed, Electrolytic (Nonsolid Electrolyte), Tantalum	.00040	1	2	4	1
CLR	39006	Capacitor, Fixed, Electrolytic (Nonsolid Electrolyte), Tantalum, Established Reliability	.00040	1	2	4	1
CRL	83500	Capacitor, Fixed, Electrolytic (Nonsolid Electrolyte), Tantalum Cathode	.00040	1	2	4	1
CU, CUR	39018	Capacitor, Fixed, Electrolytic (Aluminum Oxide), Established Reliability and Nonestablished Reliability	.00012	2	2	1	1
CE	62	Capacitor, Fixed Electrolytic (DC, Aluminum, Dry Electrolyte, Polarized)	.00012	2	2	1	1
CV	81	Capacitor, Variable, Ceramic Dielectric (Trimmer)	.0079	1	1	5	1
PC	14409	Capacitor, Variable (Piston Type, Tubular Trimmer)	.0060	2	1	5	1
CT	92	Capacitor, Variable, Air Dielectric (Trimmer)	.0000072	2	1	5	1
CG	23183	Capacitor, Fixed or Variable, Vacuum Dielectric	.0060	1	1	5	1

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10.1 CAPACITORS

Temperature Factor - π_T

T(°C)	Column 1	Column 2
20	.91	.79
30	1.1	1.3
40	1.3	1.9
50	1.6	2.9
60	1.8	4.2
70	2.2	6.0
80	2.5	8.4
90	2.8	11
100	3.2	15
110	3.7	21
120	4.1	27
130	4.6	35
140	5.1	44
150	5.6	56

$$\pi_T = \exp\left(\frac{-Ea}{8.617 \times 10^{-5}}\left(\frac{1}{T + 273} - \frac{1}{298}\right)\right)$$

Column 1: Ea = .15

Column 2: Ea = .35

T = Capacitor Ambient Temperature

NOTE: 1. π_T values shown should only be used up to the temperature rating of the device.

2. For devices with ratings higher than 150°C, use the equation to determine π_T (for applications above 150°C).

Capacitance Factor - π_C

Capacitance, C(μF)	Column 1	Column 2
.000001	.29	.04
.00001	.35	.07
.0001	.44	.12
.001	.54	.20
.01	.66	.35
.05	.76	.50
.1	.81	.59
.5	.94	.85
1	1.0	1.0
3	1.1	1.3
8	1.2	1.6
18	1.3	1.9
40	1.4	2.3
200	1.6	3.4
1000	1.9	4.9
3000	2.1	6.3
10000	2.3	8.3
30000	2.5	11
60000	2.7	13
120000	2.9	15

Column 1: $\pi_C = C^{.09}$

Column 2: $\pi_C = C^{.23}$

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10.1 CAPACITORS

Voltage Stress Factor - π_V

Voltage Stress	Column 1	Column 2	Column 3	Column 4	Column 5
0.1	1.0	1.0	1.0	1.0	1.0
0.2	1.0	1.0	1.0	1.0	1.1
0.3	1.0	1.0	1.1	1.0	1.2
0.4	1.1	1.0	1.3	1.0	1.5
0.5	1.4	1.2	1.6	1.0	2.0
0.6	2.0	2.0	2.0	2.0	2.7
0.7	3.2	5.7	2.6	15	3.7
0.8	5.2	19	3.4	130	5.1
0.9	8.6	59	4.4	990	6.8
1	14	166	5.6	5900	9.0

Column 1: $\pi_V = \left(\frac{S}{.6}\right)^5 + 1$

Column 2: $\pi_V = \left(\frac{S}{.6}\right)^{10} + 1$

Column 3: $\pi_V = \left(\frac{S}{.6}\right)^3 + 1$

Column 4: $\pi_V = \left(\frac{S}{.6}\right)^{17} + 1$

Column 5: $\pi_V = \left(\frac{S}{.5}\right)^3 + 1$

$S = \frac{\text{Operating Voltage}}{\text{Rated Voltage}}$

Note: Operating voltage is the sum of applied DC voltage and peak AC voltage.

Series Resistance Factor
(Tantalum CSR Style Capacitors Only) - π_{SR}

Circuit Resistance, CR (ohms/volt)	π_{SR}
>0.8	.66
>0.6 to 0.8	1.0
>0.4 to 0.6	1.3
>0.2 to 0.4	2.0
>0.1 to 0.2	2.7
0 to 0.1	3.3

$CR = \frac{\text{Eff. Res. Between Cap. and Pwr. Supply}}{\text{Voltage Applied to Capacitor}}$

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Quality Factor - π_Q

Quality	π_Q
Established Reliability Styles	
D	.001
C	.01
S,B	.03
R	.1
P	.3
M	1.0
L	1.5
Non-Established Reliability Capacitors (Most Two-Letter Styles)	3.0
Commercial or Unknown Screening Level	10.

NOTE: Established reliability styles are failure rate graded (D, C, S, etc.) based on life testing defined in the applicable military device specification. This category usually applies only to three-letter styles with an "R" suffix.

Environment Factor - π_E

Environment	π_E
G_B	1.0
G_F	10
G_M	20
N_S	7.0
N_U	15
A_{IC}	12
A_{IF}	15
A_{UC}	25
A_{UF}	30
A_{RW}	40
S_F	.50
M_F	20
M_L	50
C_L	570

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10.2 CAPACITORS, EXAMPLE

Example

Given: A 400 VDC rated capacitor type CQ09A1KE153K3 is being used in a fixed ground environment, 50°C component ambient temperature, and 200 VDC applied with 50 Vrms @ 60 Hz. The capacitor is being procured in full accordance with the applicable specification.

The letters "CQ" in the type designation indicate that the specification is MIL-C-19978 and that it is a Non-Established Reliability quality level. The "E" in the designation corresponds to a 400 volt DC rating. The "153" in the designation expresses the capacitance in picofarads. The first two digits are significant and the third is the number of zeros to follow. Therefore, this capacitor has a capacitance of 15,000 picofarads. (NOTE: Pico = 10^{-12}, $\mu = 10^{-6}$)

Based on the given information the following model factors are determined from the tables shown in Section 10.1.

$\lambda_b = .00051$

$\pi_T = 1.6$

$\pi_C = .69$ Use Table Equation (Note 15,000 pF = .015 μF)

$\pi_V = 2.9$ $S = \dfrac{\text{DC Volts Applied} + \sqrt{2}\,(\text{AC Volts Applied})}{\text{DC Rated Voltage}}$

$S = \dfrac{200 + \sqrt{2}\,(50)}{400} = .68$

$\pi_{SR} = 1$

$\pi_Q = 3.0$

$\pi_E = 10$

$\lambda_p = \lambda_b \pi_T \pi_C \pi_V \pi_{SR} \pi_Q \pi_E = (.00051)(1.6)(.69)(2.9)(1)(3.0)(10)$

$\lambda_p = .049$ Failures/10^6 Hours

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11.1 INDUCTIVE DEVICES, TRANSFORMERS

SPECIFICATION	STYLE	DESCRIPTION
MIL-T-27	TF	Audio, Power and High Power Pulse
MIL-T-21038	TP	Low Power Pulse
MIL-T-55631	–	Intermediate Frequency (IF), RF and Discriminator

$$\lambda_p = \lambda_b \pi_T \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - λ_b

Transformer	λ_b (F/10^6 hrs.)
Flyback (< 20 Volts)	.0054
Audio (15 -20K Hz)	.014
Low Power Pulse (Peak Pwr. < 300W, Avg. Pwr. < 5W)	.022
High Power, High Power Pulse (Peak Power ≥ 300W, Avg. Pwr. ≥ 5W)	.049
RF (10K - 10M Hz)	.13

Quality Factor - π_Q

Quality	π_Q
MIL-SPEC	1
Lower	3

Environment Factor - π_E

Environment	π_E
G_B	1.0
G_F	6.0
G_M	12
N_S	5.0
N_U	16
A_{IC}	6.0
A_{IF}	8.0
A_{UC}	7.0
A_{UF}	9.0
A_{RW}	24
S_F	.50
M_F	13
M_L	34
C_L	610

Temperature Factor - π_T

T_{HS}(°C)	π_T
20	.93
30	1.1
40	1.2
50	1.4
60	1.6
70	1.8
80	1.9
90	2.2
100	2.4
110	2.6
120	2.8
130	3.1
140	3.3
150	3.5
160	3.8
170	4.1
180	4.3
190	4.6

$$\pi_T = \exp\left(\frac{-.11}{8.617 \times 10^{-5}}\left(\frac{1}{T_{HS} + 273} - \frac{1}{298}\right)\right)$$

T_{HS} = Hot Spot Temperature (°C), See Section 11.3. This prediction model assumes that the insulation rated temperature is not exceeded for more than 5% of the time.

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11.1 INDUCTIVE DEVICES, TRANSFORMERS

Transformer Characteristic Determination Note

MIL-T-27 Example Designation



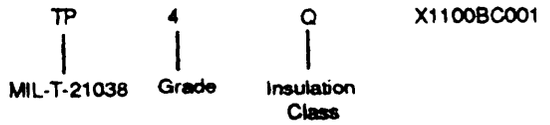
Family Type Codes Are:

Power Transformer and Filter: 01 through 09, 37 through 41

Audio Transformer: 10 through 21, 50 through 53

Pulse Transformer: 22 through 36, 54

MIL-T-21038 Example Designation



MIL-T-55631. The Transformers are Designated with the following Types, Grades and Classes.

Type I - Intermediate Frequency Transformer
Type II - Radio Frequency Transformer
Type III - Discriminator Transformer

Grade 1 - For Use When Immersion and Moisture Resistance Tests are Required
Grade 2 - For Use When Moisture Resistance Test is Required
Grade 3 - For Use in Sealed Assemblies

Class O - 85°C Maximum Operating Temperature
Class A - 105°C Maximum Operating Temperature
Class B - 125°C Maximum Operating Temperature
Class C - > 125°C Maximum Operating Temperature

The class denotes the maximum operating temperature (temperature rise plus maximum ambient temperature).

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11.2 INDUCTIVE DEVICES, COILS

SPECIFICATION	STYLE	DESCRIPTION
MIL-C-15305	-	Fixed and Variable, RF
MIL-C-83446	-	Fixed and Variable, RF, Chip
MIL-C-39010	-	Molded, RF, Est. Rel.

$$\lambda_p = \lambda_b \pi_T \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - λ_b

Inductor Type	λ_b F/10^6 hrs.
Fixed Inductor or Choke	.000030
Variable Inductor	.000050

Temperature Factor - π_T

T_{HS}(°C)	π_T
20	.93
30	1.1
40	1.2
50	1.4
60	1.6
70	1.8
80	1.9
90	2.2
100	2.4
110	2.6
120	2.8
130	3.1
140	3.3
150	3.5
160	3.8
170	4.1
180	4.3
190	4.6

$$\pi_T = \exp\left(\frac{-.11}{8.617 \times 10^{-5}}\left(\frac{1}{T_{HS} + 273} - \frac{1}{298}\right)\right)$$

T_{HS} = Hot Spot Temperature (°C), See Section 11.3

Quality Factor - π_Q

Quality	π_Q
S	.03
R	.10
P	.30
M	1.0
MIL-SPEC	1.0
Lower	3.0

Environment Factor - π_E

Environment	π_E
G_B	1.0
G_F	6.0
G_M	12
N_S	5.0
N_U	16
A_{IC}	6.0
A_{IF}	8.0
A_{UC}	7.0
A_{UF}	9.0
A_{RW}	24
S_F	.50
M_F	13
M_L	34
C_L	610

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11.3 INDUCTIVE DEVICES, DETERMINATION OF HOT SPOT TEMPERATURE

Hot Spot temperature can be estimated as follows:

$$T_{HS} = T_A + 1.1\ (\Delta T)$$

where:

T_{HS} = Hot Spot Temperature (°C)
T_A = Inductive Device Ambient Operating Temperature (°C)
ΔT = Average Temperature Rise Above Ambient (°C)

ΔT can either be determined by the appropriate "Temperature Rise" Test Method paragraph in the device base specification (e.g., paragraph 4.8.12 for MIL-T-27E), or by approximation using one of the procedures described below. For space environments a dedicated thermal analysis should be performed.

ΔT Approximation (Non-space Environments)

Information Known	ΔT Approximation
1. MIL-C-39010 Slash Sheet Number MIL-C-39010/1C-3C, 5C, 7C, 9A, 10A, 13, 14	$\Delta T = 15°C$
MIL-C-39010/4C, 6C, 8A, 11, 12	$\Delta T = 35°C$
2. Power Loss Case Radiating Surface Area	$\Delta T = 125\ W_L/A$
3. Power Loss Transformer Weight	$\Delta T = 11.5\ W_L/(Wt.)^{.6766}$
4. Input Power Transformer Weight (Assumes 80% Efficiency)	$\Delta T = 2.1\ W_I/(Wt.)^{.6766}$

W_L = Power Loss (W)

A = Radiating Surface Area of Case (in^2). See below for MIL-T-27 Case Areas

Wt. = Transformer Weight (lbs.)

W_I = Input Power (W)

NOTE: Methods are listed in preferred order (i.e., most to least accurate). MIL-C-39010 are micro-miniature devices with surface areas less than 1 in^2. Equations 2-4 are applicable to devices with surface areas from 3 in^2 to 150 in^2. Do not include the mounting surface when determining radiating surface area.

MIL-T-27 Case Radiating Areas (Excludes Mounting Surface)

Case	Area (in^2)	Case	Area (in^2)	Case	Area (in^2)
AF	4	GB	33	LB	82
AG	7	GA	43	LA	98
AH	11	HB	42	MB	98
AJ	18	HA	53	MA	115
EB	21	JB	58	NB	117
EA	23	JA	71	NA	139
FB	25	KB	72	OA	146
FA	31	KA	84		

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12.1 ROTATING DEVICES, MOTORS

The following failure-rate model applies to motors with power ratings below one horsepower. This model is applicable to polyphase, capacitor start and run and shaded pole motors. It's application may be extended to other types of fractional horsepower motors utilizing rolling element grease packed bearings. The model is dictated by two failure modes, bearing failures and winding failures. Application of the model to D.C. brush motors assumes that brushes are inspected and replaced and are not a failure mode. Typical applications include fans and blowers as well as various other motor applications. The model is based on References 4 and 37, which contain a more comprehensive treatment of motor life prediction methods. The references should be reviewed when bearing loads exceed 10 percent of rated load, speeds exceed 24,000 rpm or motor loads include motor speed slip of greater than 25 percent.

The instantaneous failure rates, or hazard rates, experienced by motors are not constant but increase with time. The failure rate model in this section is an average failure rate for the motor operating over time period "t". This time period is either the system design life cycle (LC) or the time period the motor must last between complete refurbishment (or replacement). The model assumes that motors are replaced upon failure and that an effective constant failure rate is achieved after a given time due to the fact that the effective "time zero" of replaced motors becomes random after a significant portion of the population is replaced. The average failure rate, λ_p, can be treated as a constant failure rate and added to other part failure rates from this Handbook.

$$\lambda_p = \left[\frac{\lambda_1}{A\alpha_B} + \frac{\lambda_2}{B\alpha_W} \right] \times 10^6 \text{ Failures}/10^6 \text{ Hours}$$

Bearing & Winding Characteristic Life - α_B and α_W

T_A (°C)	α_B (Hr.)	α_W (Hr.)	T_A (°C)	α_B (Hr.)	α_W (Hr.)
0	3600	6.4e+06	70	22000	1.1e+05
10	13000	3.2e+06	80	14000	7.0e+04
20	39000	1.6e+06	90	9100	4.6e+04
30	78000	8.9e+05	100	6100	3.1e+04
40	80000	5.0e+05	110	4200	2.1e+04
50	55000	2.9e+05	120	2900	1.5e+04
60	35000	1.8e+05	130	2100	1.0e+04
			140	1500	7.5e+03

$$\alpha_B = \left[10^{\left(2.534 - \frac{2357}{T_A + 273}\right)} + \frac{1}{10^{\left(20 - \frac{4500}{T_A + 273}\right)} + 300} \right]^{-1}$$

$$\alpha_W = 10^{\left[\frac{2357}{T_A + 273} - 1.83\right]}$$

α_B = Weibull Characteristic Life for the Motor Bearing

α_W = Weibull Characteristic Life for the Motor Windings

T_A = Ambient Temperature (°C)

NOTE: See page 12-3 for method to calculate α_B and α_W when temperature is not constant.

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12.1 ROTATING DEVICES, MOTORS

A and B Determination

Motor Type	A	B
Electrical (General)	1.9	1.1
Sensor	.48	.29
Servo	2.4	1.7
Stepper	11	5.4

λ_1 and λ_2 Determination

$\frac{LC}{\alpha B}$ or $\frac{LC}{\alpha_W}$	λ_1 or λ_2
0 - .10	.13
.11 - .20	.15
.21 - .30	.23
.31 - .40	.31
.41 - .50	.41
.51 - .60	.51
.61 - .70	.61
.71 - .80	.68
.81 - .90	.76
> 1.0	1.0

LC is the system design life cycle (in hours), or the motor preventive maintenance interval, if motors will be periodically replaced or refurbished. Determine λ_1 and λ_2 separately based on the respective $\frac{LC}{\alpha_B}$ and $\frac{LC}{\alpha_W}$ ratios.

Example Calculation

A general purpose electrical motor is operating at 50°C in a system with a 10 year design life (87600 hours) expectancy,

α_B = 55000 Hrs.

α_W = 2.9e + 5 Hrs.

$$\frac{LC}{\alpha_B} = \frac{87600 \text{ Hrs.}}{55000 \text{ Hrs.}} = 1.6$$

$$\frac{LC}{\alpha_W} = \frac{87600 \text{ Hrs.}}{2.9e + 5 \text{ Hrs.}} = .3$$

λ_1 = 1.0 $\left(\text{for } \frac{LC}{\alpha_B} = 1.6\right)$

λ_2 = .23 $\left(\text{for } \frac{LC}{\alpha_W} = .3\right)$

A = 1.9

B = 1.1

$$\lambda_p = \left[\frac{1.0}{(1.9)(55000)} + \frac{.23}{(1.1)(2.9e + 5)}\right] \times 10^6$$

λ_p = 10.3 Failures/10^6 Hours

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$\alpha_{Calculation}$ for Cycled Temperature

The following equation can be used to calculate a weighted characteristic life for both bearings and windings (e.g., for bearings substitute α_B for all α's in equation).

$$\alpha = \frac{(h_1 + h_2 + h_3 + \cdots\cdots h_m)}{\frac{h_1}{\alpha_1} + \frac{h_2}{\alpha_2} + \frac{h_3}{\alpha_3} + \cdots\cdots \frac{h_m}{\alpha_m}}$$

where:

α = either α_B or α_W

h_1 = Time at Temperature T_1

h_2 = Time to Cycle From Temperature T_1 to T_3

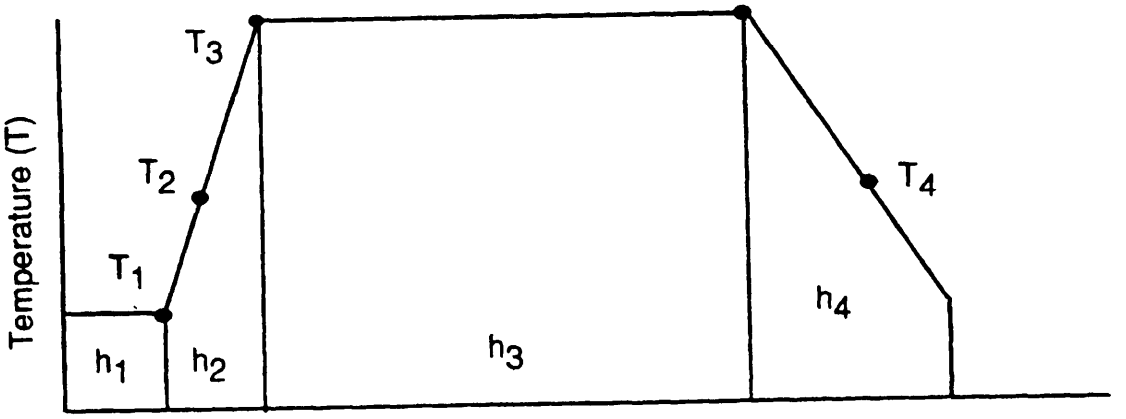
h_3 = Time at Temperature T_3

h_m = Time at Temperature T_m

α_1 = Bearing (or Winding) Life at T_1

α_2 = Bearing (or Winding) Life at T_2

NOTE: $T_2 = \frac{T_1 + T_3}{2}$, $T_4 = \frac{T_3 + T_1}{2}$



Thermal Cycle

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12.2 ROTATING DEVICES, SYNCHROS AND RESOLVERS

DESCRIPTION
Rotating Synchros and Resolvers

$$\lambda_p = \lambda_b \pi_S \pi_N \pi_E \text{ Failures/}10^6 \text{ Hours}$$

NOTE: Synchros and resolvers are predominately used in service requiring only slow and infrequent motion. Mechanical wearout problems are infrequent so that the electrical failure mode dominates, and no mechanical mode failure rate is required in the model above.

Base Failure Rate - λ_b

T_F (°C)	λ_b	T_F (°C)	λ_b
30	.0083	85	.032
35	.0088	90	.041
40	.0095	95	.052
45	.010	100	.069
50	.011	105	.094
55	.013	110	.13
60	.014	115	.19
65	.016	120	.29
70	.019	125	.45
75	.022	130	.74
80	.027	135	1.3

$$\lambda_b = .00535 \exp\left(\frac{T_F + 273}{334}\right)^{8.5}$$

T_F = Frame Temperature (°C)

If Frame Temperature is Unknown Assume
T_F = 40 °C + Ambient Temperature

Size Factor - π_S

DEVICE TYPE	π_S		
	Size 8 or Smaller	Size 10-16	Size 18 or Larger
Synchro	2	1.5	1
Resolver	3	2.25	1.5

Number of Brushes Factor - π_N

Number of Brushes	π_N
≤ 2	1.4
3	2.5
4	3.2

Environment Factor - π_E

Environment	π_E
G_B	1.0
G_F	2.0
G_M	12
N_S	7.0
N_U	18
A_{IC}	4.0
A_{IF}	6.0
A_{UC}	16
A_{UF}	25
A_{RW}	26
S_F	.50
M_F	14
M_L	36
C_L	680

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12.3 ROTATING DEVICES, ELAPSED TIME METERS

DESCRIPTION
Elapsed Time Meters

$$\lambda_p = \lambda_b \pi_T \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - λ_b

Type	λ_b
A.C.	20
Inverter Driven	30
Commutator D.C.	80

Temperature Stress Factor - π_T

Operating T (°C)/Rated T (°C)	π_T
0 to .5	.5
.6	.6
.8	.8
1.0	1.0

Environment Factor - π_E

Environment	π_E
G_B	1.0
G_F	2.0
G_M	12
N_S	7.0
N_U	18
A_{IC}	5.0
A_{IF}	8.0
A_{UC}	16
A_{UF}	25
A_{RW}	26
S_F	.50
M_F	14
M_L	38
C_L	N/A

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13.1 RELAYS, MECHANICAL

SPECIFICATION
MIL-R-5757
MIL-R-6106
MIL-R-13718
MIL-R-19648
MIL-R-19523
MIL-R-39016
MIL-R-83516
MIL-R-83520
MIL-R-83536
MIL-R-83725
MIL-R-83726 (Except Class C, Solid State Type)

DESCRIPTION
Mechanical Relay

$$\lambda_p = \lambda_b \pi_L \pi_C \pi_{CYC} \pi_F \pi_Q \pi_E \text{ Failures}/10^6 \text{ Hours}$$

Base Failure Rate - λ_b

T_A (°C)	Rated Temperature 85°C[1]	125°C[2]
25	.0059	.0059
30	.0067	.0066
35	.0075	.0073
40	.0084	.0081
45	.0094	.0089
50	.010	.0098
55	.012	.011
60	.013	.012
65	.014	.013
70	.016	.014
75	.017	.015
80	.019	.017
85	.021	.018
90		.019
95		.021
100		.022
105		.024
110		.026
115		.027
120		.029
125		.031

1. $\lambda_b = .0059 \exp\left(\frac{-.19}{8.617 \times 10^{-5}}\left[\frac{1}{T + 273} - \frac{1}{298}\right]\right)$

2. $\lambda_b = .0059 \exp\left(\frac{-.17}{8.617 \times 10^{-5}}\left[\frac{1}{T + 273} - \frac{1}{298}\right]\right)$

T_A = Ambient Temperature (°C)

Load Stress Factor - π_L

S	Load Type Resistive[1]	Inductive[2]	Lamp[3]
.05	1.00	1.02	1.06
.10	1.02	1.06	1.28
.20	1.06	1.28	2.72
.30	1.15	1.76	9.49
.40	1.28	2.72	54.6
.50	1.48	4.77	
.60	1.76	9.49	
.70	2.15	21.4	
.80	2.72		
.90	3.55		
1.00	4.77		

1. $\pi_L = \exp\left(\frac{S}{.8}\right)^2$

2. $\pi_L = \exp\left(\frac{S}{.4}\right)^2$

3. $\pi_L = \exp\left(\frac{S}{.2}\right)^2$

$S = \frac{\text{Operating Load Current}}{\text{Rated Resistive Load Current}}$

For single devices which switch two different load types, evaluate π_L for each possible stress load type combination and use the worse case (largest π_L).

Contact Form Factor - π_C

(Applies to Active Conducting Contacts)

Contact Form	π_C
SPST	1.00
DPST	1.50
SPDT	1.75
3PST	2.00
4PST	2.50
DPDT	3.00
3PDT	4.25
4PDT	5.50
6PDT	8.00

Cycling Factor - π_{CYC}

Cycle Rate (Cycles per Hour)	π_{CYC} (MIL-SPEC)
≥ 1.0	$\frac{\text{Cycles per Hour}}{10}$
< 1.0	0.1

Cycle Rate (Cycles per Hour)	π_{CYC} (Commercial Quality)
> 1000	$\left(\frac{\text{Cycles per Hour}}{100}\right)^2$
10 - 1000	$\frac{\text{Cycles per Hour}}{10}$
< 10	1.0

NOTE: Values of π_{CYC} for cycling rates beyond the basic design limitations of the relay are not valid. Design specifications should be consulted prior to evaluation of π_{CYC}.

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13.1 RELAYS, MECHANICAL

Quality Factor - π_Q

Quality	π_Q
R	.10
P	.30
X	.45
U	.60
M	1.0
L	1.5
MIL-SPEC, Non-Est. Rel.	1.5
Commercial	2.9

Environment Factor - π_E

Environment	π_E
G_B	1.0
G_F	2.0
G_M	15
N_S	8.0
N_U	27
A_{IC}	7.0
A_{IF}	9.0
A_{UC}	11
A_{UF}	12
A_{RW}	46
S_F	.50
M_F	25
M_L	66
C_L	N/A

Application and Construction Factor - π_F

Contact Rating	Application Type	Construction Type	π_F
Signal Current (Low mv and ma)	Dry Circuit	Armature (Long)	4
		Dry Reed	6
		Mercury Wetted	1
		Magnetic Latching	4
		Balanced Armature	7
		Solenoid	7
0-5 Amp	General Purpose	Armature (Long)	3
		Balanced Armature	5
		Solenoid	6
	Sensitive (0 - 100 mw)	Armature (Long and Short)	5
		Mercury Wetted	2
		Magnetic Latching	6
		Meter Movement	100
		Balanced Armature	10
	Polarized	Armature (Short)	10
		Meter Movement	100
	Vibrating Reed	Dry Reed	6
		Mercury Wetted	1
	High Speed	Armature (Balanced and Short)	25
		Dry Reed	6
	Thermal Time Delay	Bimetal	10
	Electronic Time Delay, Non-Thermal		9
	Latching, Magnetic	Dry Reed	10
		Mercury Wetted	5
		Balanced armature	5
5-20 Amp	High Voltage	Vacuum (Glass)	20
		Vacuum (Ceramic)	5
	Medium Power	Armature (Long and Short)	3
		Mercury Wetted	1
		Magnetic Latching	2
		Mechanical Latching	3
		Balanced Armature	2
		Solenoid	2
25-600 Amp	Contactors (High Current)	Armature (Short)	7
		Mechanical Latching	12
		Balanced Armature	10
		Solenoid	5

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Supersedes page 13-2 of Revision F

13.2 RELAYS, SOLID STATE AND TIME DELAY

SPECIFICATION
MIL-R-28750
MIL-R-83726

DESCRIPTION
Relay, Solid State
Relay, Time Delay, Hybrid and Solid State

The most accurate method for predicting the failure rate of solid state (and solid state time delay) relays is to sum the failure rates for the individual components which make up the relay. The individual component failure rates can either be calculated from the models provided in the main body of this Handbook (Parts Stress Method) or from the Parts Count Method shown in Appendix A, depending upon the depth of knowledge the analyst has about the components being used. If insufficient information is available, the following default model can be used:

$$\lambda_p = \lambda_b \pi_Q \pi_E \text{ Failures}/10^6 \text{ Hours}$$

Base Failure Rate - λ_b

Relay Type	λ_b
Solid State	.029
Solid State Time Delay	.029
Hybrid	.029

Quality Factor - π_Q

Quality	π_Q
MIL-SPEC	1.0
Commercial	1.9

Environment Factor - π_E

Environment	π_E
G_B	1.0
G_F	3.0
G_M	12
N_S	6.0
N_U	17
A_{IC}	12
A_{IF}	19
A_{UC}	21
A_{UF}	32
A_{RW}	23
S_F	.40
M_F	12
M_L	33
C_L	590

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$$\lambda_p = \lambda_b \pi_L \pi_C \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - λ_b

Description	Spec. MIL-S-	λ_b (F/10^6 Hrs.)
Centrifugal	N/A	3.4
Dual-In-line Package	83504	.00012
Limit	8805	4.3
Liquid Level	21277	2.3
Microwave (Waveguide)	N/A	1.7
Pressure	8932	2.8
	9395	
Pushbutton	1211	.10
	8805	
	22885	
	24317	
Reed	55433	.0010
Rocker	3950	.023
	22885	
Rotary	3786	.11
	13623	
	15291	
	15743	
	22604	
	22710	
	45885	
	82359	
Sensitive	8805	.49
	13484	
	22614	
Thermal	12285	.031
	24286	
Thumbwheel	22710	.18
Toggle	3950	.10
	5594	
	8805	
	8834	
	9419	
	13735	
	81551	
	83731	

Load Stress Factor - π_L

Stress S	Load Type		
	Resistive	Inductive	Lamp
0.05	1.00	1.02	1.06
0.1	1.02	1.06	1.28
0.2	1.06	1.28	2.72
0.3	1.15	1.76	9.49
0.4	1.28	2.72	54.6
0.5	1.48	4.77	
0.6	1.76	9.49	
0.7	2.15	21.4	
0.8	2.72		
0.9	3.55		
1.0	4.77		

$$S = \frac{\text{Operating Load Current}}{\text{Rated Resistive Load Current}}$$

$\pi_L = \exp (S/.8)^2$ for Resistive Load

$\pi_L = \exp (S/.4)^2$ for Inductive Load

$\pi_L = \exp (S/.2)^2$ for Lamp Load

NOTE: When the switch is rated by inductive load, then use resistive π_L.

Contact Configuration Factor* - π_C

Contact Form	# of Contacts, NC	π_C
SPST	1	1.0
DPST	2	1.3
SPDT	2	1.3
3PST	3	1.4
4PST	4	1.6
DPDT	4	1.6
3PDT	6	1.8
4PDT	8	2.0
6PDT	12	2.3

$\pi_C = (NC)^{.33}$

* Applies to toggle and pushbutton switches only, all others use $\pi_C = 1$.

Supersedes page 14-1 through 14-4 of Revision F

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14.1 SWITCHES

Quality Factor - π_Q

Quality	π_Q
MIL-SPEC	1
Lower	2

Environment Factor - π_E

Environment	π_E
G_B	1.0
G_F	3.0
G_M	18
N_S	8.0
N_U	29
A_{IC}	10
A_{IF}	18
A_{UC}	13
A_{UF}	22
A_{RW}	46
S_F	.50
M_F	25
M_L	67
C_L	1200

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14.2 SWITCHES, CIRCUIT BREAKERS

SPECIFICATION	DESCRIPTION
MIL-C-13516	Circuit Breakers, Manual and Automatic
MIL-C-55629	Circuit Breakers, Magnetic, Unsealed, Trip-Free
MIL-C-83383	Circuit Breakers, Remote Control, Thermal, Trip-Free
MIL-C-39019	Circuit Breakers, Magnetic, Low Power, Sealed, Trip-Free Service
W-C-375	Circuit Breakers, Molded Case, Branch Circuit and Service

$$\lambda_p = \lambda_b \pi_C \pi_U \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - λ_b

Description	λ_b
Magnetic	.34
Thermal	.34
Thermal-Magnetic	.34

Configuration Factor - π_C

Configuration	π_C
SPST	1.0
DPST	2.0
3PST	3.0
4PST	4.0

Use Factor - π_U

Use	π_U
Not Used as a Power On/Off Switch	1.0
Also Used as a Power On/Off Switch	2.5

Quality Factor - π_Q

Quality	π_Q
MIL-SPEC	1.0
Lower	8.4

Environment Factor - π_E

Environment	π_E
G_B	1.0
G_F	2.0
G_M	15
N_S	8.0
N_U	27
A_{IC}	7.0
A_{IF}	9.0
A_{UC}	11
A_{UF}	12
A_{RW}	46
S_F	.50
M_F	25
M_L	66
C_L	N/A

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15.1 CONNECTORS, GENERAL

$$\lambda_p = \lambda_b \pi_T \pi_K \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

APPLICATION NOTE: The failure rate model is for a mated pair of connectors. It is sometimes desirable to assign half of the overall mated pair connector (i.e., single connector) failure rate to the line replaceable unit and half to the chassis (or backplane). An example of when this would be beneficial is for input to maintainability prediction to allow a failure rate weighted repair time to be estimated for both the LRU and chassis. This accounting procedure could be significant if repair times for the two halves of the connector are substantially different. For a single connector divide λ_p by two.

Base Failure Rate - λ_b

Description	Specification MIL-C-	λ_b
Circular/Cylindrical	5015 26482 26500 27599 28840 29600 38999 83723 81511	.0010
Card Edge (PCB)*	21097 55302	.040
Hexagonal	24055 24056	.15
Rack and Panel	24308 28731 28748 83515	.021
Rectangular	21617 24308 28748 28804 81659 83513 83527 83733 85028	.046
RF Coaxial	3607 15370 3643 25516 3650 26637 3655 39012 55235 83517	.00041
Telephone	55074	.0075
Power	22992	.0070
Triaxial	49142	.0036

Temperature Factor - π_T

T_o (°C)	π_T
20	.91
30	1.1
40	1.3
50	1.5
60	1.8
70	2.0
80	2.3
90	2.7
100	3.0
110	3.4
120	3.7
130	4.1
140	4.6
150	5.0
160	5.5
170	6.0
180	6.5
190	7.0
200	7.5
210	8.1
220	8.6
230	9.2
240	9.8
250	10.

$$\pi_T = \exp\left[\frac{-.14}{8.617 \times 10^{-5}}\left(\frac{1}{T_o + 273} - \frac{1}{298}\right)\right]$$

T_o = Connector Ambient + ΔT

ΔT = Connector Insert Temperature Rise (See Table)

* Printed Circuit Board Connector

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15.1 CONNECTORS, GENERAL

Default Insert Temperature Rise (ΔT °C) Determination

Amperes Per Contact	Contact Gauge 30	22	20	16	12
2	10	4	2	1	0
3	22	8	5	2	1
4	37	13	8	4	1
5	56	19	13	5	2
6	79	27	18	8	3
7		36	23	10	4
8		46	30	13	5
9		57	37	16	6
10		70	45	19	7
15			96	41	15
20				70	26
25				106	39
30					54
35					72
40					92

$\Delta T = 3.256\ (i)^{1.85}$ 32 Gauge Contacts
$\Delta T = 2.856\ (i)^{1.85}$ 30 Gauge Contacts
$\Delta T = 2.286\ (i)^{1.85}$ 28 Gauge Contacts
$\Delta T = 1.345\ (i)^{1.85}$ 24 Gauge Contacts
$\Delta T = 0.989\ (i)^{1.85}$ 22 Gauge Contacts
$\Delta T = 0.640\ (i)^{1.85}$ 20 Gauge Contacts
$\Delta T = 0.429\ (i)^{1.85}$ 18 Gauge Contacts
$\Delta T = 0.274\ (i)^{1.85}$ 16 Gauge Contacts
$\Delta T = 0.100\ (i)^{1.85}$ 12 Gauge Contacts

ΔT = Insert Temperature Rise
i = Amperes per Contact

RF Coaxial Connectors $\Delta T = 5°C$

RF Coaxial Connectors (High Power Applications) $\Delta T = 50°C$

Mating/Unmating Factor - π_K

Mating/Unmating Cycles* (per 1000 hours)	π_K
0 to .05	1.0
> .05 to .5	1.5
> .5 to 5	2.0
> 5 to 50	3.0
> 50	4.0

*One cycle includes both connect and disconnect.

Quality Factor - π_Q

Quality	π_Q
MIL-SPEC	1
Lower	2

Environment Factor - π_E

Environment	π_E
G_B	1.0
G_F	1.0
G_M	8.0
N_S	5.0
N_U	13
A_{IC}	3.0
A_{IF}	5.0
A_{UC}	8.0
A_{UF}	12
A_{RW}	19
S_F	.50
M_F	10
M_L	27
C_L	490

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15.2 CONNECTORS, SOCKETS

$$\lambda_p = \lambda_b \pi_P \pi_Q \pi_E \quad \text{Failures/}10^6 \text{ Hours}$$

Base Failure Rate - λ_b

Description	Spec. MIL-S	λ_b
Dual-In-Line Package	83734	.00064
Single-In-Line Package	83734	.00064
Chip Carrier	38533	.00064
Pin Grid Array	N/A	.00064
Relay	12883	.037
Transistor	12883	.0051
Electron Tube, CRT	12883	.011

Quality Factor - π_Q

Quality	π_Q
MIL-SPEC.	.3
Lower	1.0

Environment Factor - π_E

Environment	π_E
G_B	1.0
G_F	3.0
G_M	14
N_S	6.0
N_U	18
A_{IC}	8.0
A_{IF}	12
A_{UC}	11
A_{UF}	13
A_{RW}	25
S_F	.50
M_F	14
M_L	36
C_L	650

Active Pins Factor - π_P

Number of Active Contacts	π_P	Number of Active Contacts	π_P
1	1.0	55	6.9
2	1.5	60	7.4
3	1.7	65	7.9
4	1.9	70	8.4
5	2.0	75	8.9
6	2.1	80	9.4
7	2.3	85	9.9
8	2.4	90	10
9	2.5	95	11
10	2.6	100	12
11	2.7	105	12
12	2.8	110	13
13	2.9	115	13
14	3.0	120	14
15	3.1	125	14
16	3.2	130	15
17	3.3	135	16
18	3.4	140	16
19	3.5	145	17
20	3.6	150	18
25	4.1	155	18
30	4.5	160	19
35	5.0	165	20
40	5.5	170	20
45	5.9	175	21
50	6.4	180	22

$$\pi_P = \exp\left(\frac{N-1}{10}\right)^q$$

$q = .39$

N = Number of Active Pins

An active contact is the conductive element which mates with another element for the purpose of transferring electrical energy.

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16.1 INTERCONNECTION ASSEMBLIES WITH PLATED THROUGH HOLES

$$\lambda_p = \lambda_b\left[N_1\,\pi_C + N_2\,(\pi_C + 13)\right]\pi_Q\pi_E \text{ Failures}/10^6 \text{ Hours}$$

APPLICATION NOTE: This model applies to board configurations with leaded devices mounted into the plated through holes and assumes failures are predominately defect related. For boards using surface mount technology, use Section 16.2. For a mix of leaded devices mounted into plated through holes and surface mount devices, use this model for the leaded devices and use Section 16.2 for the surface mount contribution.

A discrete wiring assembly with electroless deposit plated through holes is basically a pattern of insulated wires laid down on an adhesive coated substrate. The primary cause of failure for both printed wiring and discrete wiring assemblies is associated with plated through-hole (PTH) problems (e.g., barrel cracking).

Base Failure Rate - λ_b

Technology	λ_b
Printed Wiring Assembly/Printed Circuit Boards with PTHs	.000017
Discrete Wiring with Electroless Deposited PTH (≤ 2 Levels of Circuitry)	.00011

Quality Factor - π_Q

Quality	π_Q
MIL-SPEC or Comparable Institute for Interconnecting, and Packaging Electronic Circuits (IPC) Standards (IPC Level 3)	1
Lower	2

Number of PTHs Factor - N_1 and N_2

Factor	Quantity
N_1	Automated Techniques: Quantity of Wave Infrared (IR) or Vapor Phase Soldered Functional PTHs
N_2	Quantity of Hand Soldered PTHs

Environment Factor - π_E

Environment	π_E
G_B	1.0
G_F	2.0
G_M	7.0
N_S	5.0
N_U	13
A_{IC}	5.0
A_{IF}	8.0
A_{UC}	16
A_{UF}	28
A_{RW}	19
S_F	.50
M_F	10
M_L	27
C_L	500

Complexity Factor - π_C

Number of Circuit Planes, P	π_C
≤ 2	1.0
3	1.3
4	1.6
5	1.8
6	2.0
7	2.2
8	2.4
9	2.6
10	2.8
11	2.9
12	3.1
13	3.3
14	3.4
15	3.6
16	3.7
17	3.9
18	4.0
Discrete Wiring w/PTH	1
$\pi_C = .65\,P^{.63}$	$2 \le P \le 18$

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16.2 INTERCONNECTION ASSEMBLIES, SURFACE MOUNT TECHNOLOGY

APPLICATION NOTE: The SMT Model was developed to assess the life integrity of leadless and leaded devices. It provides a relative measure of circuit card wearout due to thermal cycling fatigue failure of the "weakest link" SMT device. An analysis should be performed on all circuit board SMT components. The component with the largest failure rate value (weakest link) is assessed as the overall board failure rate due to SMT. The model assumes the board is completely renewed upon failure of the weakest link and the results do not consider solder or lead manufacturing defects. This model is based on the techniques developed in Reference 37.

λ_{SMT} = Average failure rate over the expected equipment life cycle due to surface mount device wearout. This failure rate contribution to the system is for the Surface Mount Device on each board exhibiting the highest **absolute** value of the strain range:

$$\lambda_{SMT} = \frac{ECF}{\alpha_{SMT}}$$

ECF = Effective cumulative number of failures over the Weibull characteristic life.

Effective Cumulative Failures - ECF

$\frac{LC}{\alpha_{SMT}}$	ECF
0 - .1	.13
.11 - .20	.15
.21 - .30	.23
.31 - .40	.31
.41 - .50	.41
.51 - .60	.51
.61 - .70	.61
.71 - .80	.68
.81 - .90	.76
> .9	1.0

LC = Design life cycle of the equipment in which the circuit board is operating.

α_{SMT} = The Weibull characteristic life. α_{SMT} is a function of device and substrate material, the manufacturing methods, and the application environment used.

$$\alpha_{SMT} = \frac{N_f}{CR}$$

where:

CR = Temperature cycling rate in cycles per calendar hour. Base on a thermal analysis of the circuit board. Use table default values if other estimates do not exist.

N_f = Average number of thermal cycles to failure

$$N_f = 3.5 \left(\frac{d}{.65h} \left| (\alpha_S \Delta T - \alpha_{CC} (\Delta T + T_{RISE})) \right| \times 10^{-6} \right)^{-2.26} (\pi_{LC})$$

where:

d = Distance from center of device to the furthest solder joint in mils (thousandths of an inch)

h = Solder joint height in mils for leadless devices. Default to h = 8 for all leaded configurations.

α_S = Circuit board substrate thermal coefficient of expansion (TCE)

ΔT = Use environment temperature extreme difference

α_{CC} = Package material thermal coefficient of expansion (TCE)

T_{RISE} = Temperature rise due to power dissipation (Pd)

Pd = $\theta_{JC}P$

θ_{JC} = Thermal resistance °/Watt

P = Power Dissipation (Watts)

π_{LC} = Lead configuration factor

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16.2 INTERCONNECTION ASSEMBLIES, SURFACE MOUNT TECHNOLOGY

CR - Cycling Rate Default Values

Equipment Type	Number of Cycles/Hour
Automotive	1.0
Consumer (television, radio, recorder)	.08
Computer	.17
Telecommunications	.0042
Commercial Aircraft	.25
Industrial	.021
Military Ground Applications	.03
Military Aircraft (Cargo)	.12
Military Aircraft (Fighter)	.5

π_{LC} - Lead Configuration Factor

Lead Configuration	π_{LC}
Leadless	1
J or S Lead	150
Gull Wing	5,000

α_{CC} - TCE Package Values

Substrate Material	α_{CC} Average Value
Plastic	7
Ceramic	6

ΔT - Use Environment Default Temperature Difference

Environment	ΔT
G_B	7
G_F	21
G_M	26
N_S	26
N_U	61
A_{IC}	31
A_{IF}	31
A_{UC}	57
A_{UF}	57
A_{RW}	31
S_F	7
M_F	N/A
M_L	N/A
C_L	N/A

α_S - Default TCE Substrate Values

Substrate Material	α_S
FR-4 Laminate	18
FR-4 Multilayer Board	20
FR-4 Multilayer Board w/Copper Clad Invar	11
Ceramic Multilayer Board	7
Copper Clad Invar	5
Copper Clad Molybdenum	5
Carbon-Fiber/Epoxy Composite	1
Kevlar Fiber	3
Quartz Fiber	1
Glass Fiber	5
Epoxy/Glass Laminate	15
Polyamide/Glass Laminate	13
Polyamide/Kevlar Laminate	6
Polyamide/Quartz Laminate	8
Epoxy/Kevlar Laminate	7
Alumina (Ceramic)	7
Epoxy Aramid Fiber	7
Polyamide Aramid Fiber	6
Epoxy-Quartz	9
Fiberglass Teflon Laminates	20
Porcelainized Copper Clad Invar	7
Fiberglass Ceramic Fiber	7

EXAMPLE: A large plastic encapsulated leadless chip carrier is mounted on a epoxy-glass printed wiring assembly. The design considerations are: a square package is 1480 mils on a side, solder height is 5 mils, power dissipation is .5 watts, thermal resistance is 20°C/watt, the design life is 20 years and environment is military ground application. The failure rate developed is the impact of SMT for a single circuit board and accounts for all SMT devices on this board. This failure rate is added to the sum of all of the component failure rates on the circuit board.

$$\lambda_{SMT} = \frac{ECF}{\alpha_{SMT}}$$

$$\alpha_{SMT} = \frac{N_I}{CR}$$

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16.2 INTERCONNECTION ASSEMBLIES, SURFACE MOUNT TECHNOLOGY

$$N_f = 3.5 \left(\frac{d}{(.65)(h)} \left| (\alpha_S \Delta T - \alpha_{CC} (\Delta T + T_{RISE})) \right| \times 10^{-6} \right)^{-2.26} (\pi_{LC})$$

For d: $d = \frac{1}{2}(1480) = 740$ mils

For h: $h = 5$ mils

For α_S: $\alpha_S = 15$ (Table - Epoxy Glass)

For ΔT: $\Delta T = 21$ (Table - G_F)

For α_{CC}: $\alpha_{CC} = 7$ (Table - Plastic)

For T_{RISE}: $T_{RISE} = \theta_{JC} P = 20(.5) = 10°C$

For π_{LC}: $\pi_{LC} = 1$ (Table - Leadless)

For CR: CR = .03 cycles/hour (Table - Military Ground)

$$N_f = 3.5 \left(\frac{740}{(.65)(5)} \left| (15(21) - 7(21+10)) \right| \times 10^{-6} \right)^{-2.26} \quad (1)$$

N_f = 18,893 thermal cycles to failure

$$\alpha_{SMT} = \frac{18{,}893 \text{ cycles}}{.03 \text{ cyles/hour}} = 629{,}767 \text{ hours}$$

$$\frac{LC}{\alpha_{SMT}} = \frac{(20 \text{ yrs.}) \left(8760 \frac{hr}{yr} \right)}{629{,}767 \text{ hrs.}} = .28$$

ECF = .23 failures (Table - Effective Cumulative Failures)

$$\lambda_{SMT} = \frac{ECF}{\alpha_{SMT}} = \frac{.23 \text{ failures}}{629{,}767 \text{ hours}} = .0000004 \text{ failures/hour}$$

$\lambda_{SMT} = .4$ failures/10^6 hours

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17.1 CONNECTIONS

APPLICATION NOTE: The failure rate model in this section applies to connections used on all assemblies except those using plated through holes or surface mount technology. Use the Interconnection Assembly Model in Section 16 to account for connections to a circuit board using either plated through hole technology or surface mount technology. The failure rate of the structure which supports the connections and parts, e.g., non-plated-through hole boards and terminal straps, is considered to be zero. Solderless wrap connections are characterized by solid wire wrapped under tension around a post, whereas hand soldering with wrapping does not depend on a tension induced connection. The following model is for a single connection.

$$\lambda_p = \lambda_b \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - λ_b

Connection Type	λ_b (F/10^6 hrs)
Hand Solder, w/o Wrapping	.0013
Hand Solder, w/Wrapping	.000070
Crimp	.00026
Weld	.000015
Solderless Wrap	.0000068
Clip Termination	.00012
Reflow Solder	.000069
Spring Contact	.17
Terminal Block	.062

Environment Factor - π_E

Environment	π_E
G_B	1.0
G_F	2.0
G_M	7.0
N_S	4.0
N_U	11
A_{IC}	4.0
A_{IF}	6.0
A_{UC}	6.0
A_{UF}	8.0
A_{RW}	16
S_F	.50
M_F	9.0
M_L	24
C_L	420

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APPENDIX A: PARTS COUNT RELIABILITY PREDICTION

Parts Count Reliability Prediction - This prediction method is applicable during bid proposal and early design phases when insufficient information is available to use the part stress analysis models shown in the main body of this Handbook. The information needed to apply the method is (1) generic part types (including complexity for microcircuits) and quantities, (2) part quality levels, and (3) equipment environment. The equipment failure rate is obtained by looking up a generic failure rate in one of the following tables, multiplying it by a quality factor, and then summing it with failure rates obtained for other components in the equipment. The general mathematical expression for equipment failure rate with this method is:

$$\lambda_{EQUIP} = \sum_{i=1}^{i=n} N_i (\lambda_g \pi_Q)_i \qquad \text{Equation 1}$$

for a given equipment environment where:

λ_{EQUIP} = Total equipment failure rate (Failures/10^6 Hours)

λ_g = Generic failure rate for the ith generic part (Failures/10^6 Hours)

π_Q = Quality factor for the ith generic part

N_i = Quantity of ith generic part

n = Number of different generic part categories in the equipment

Equation 1 applies if the entire equipment is being used in one environment. If the equipment comprises several units operating in different environments (such as avionics systems with units in airborne inhabited (A_I) and uninhabited (A_U) environments), then Equation 1 should be applied to the portions of the equipment in each environment. These "environment-equipment" failure rates should be added to determine total equipment failure rate. Environmental symbols are defined in Section 3.

The quality factors to be used with each part type are shown with the applicable λ_g tables and are not necessarily the same values that are used in the Part Stress Analysis. Microcircuits have an additional multiplying factor, π_L, which accounts for the maturity of the manufacturing process. For devices in production two years or more, no modification is needed. For those in production less than two years, λ_g should be multiplied by the appropriate π_L factor (See page A-4).

It should be noted that no generic failure rates are shown for hybrid microcircuits. Each hybrid is a fairly unique device. Since none of these devices have been standardized, their complexity cannot be determined from their name or function. Identically or similarly named hybrids can have a wide range of complexity that thwarts categorization for purposes of this prediction method. If hybrids are anticipated for a design, their use and construction should be thoroughly investigated on an individual basis with application of the prediction model in Section 5.

The failure rates shown in this Appendix were calculated by assigning model default values to the failure rate models of Section 5 through 23. The specific default values used for the model parameters are shown with the λ_g Tables for microcircuits. Default parameters for all other part classes are summarized in the tables starting on Page A-12. For parts with characteristics which differ significantly from the assumed defaults, or parts used in large quantities, the underlying models in the main body of this Handbook can be used.

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APPENDIX A: PARTS COUNT

Generic Failure Rate, λ_g (Failures/10^6 Hours) for Microcircuits. See Page A-4 for π_Q Values

(Defaults: π_T Based on Ea Shown, Solder or Weld Seal DIPs/PGAs (No. Pins as Shown Below), π_L = 1 (Device in Production ≥ 2 Yr.))

Section #	Part Type	Environ. → / T_J (°C) →	G_B 50	G_F 60	G_M 65	N_S 60	N_U 65	A_{IC} 75	A_{IF} 75	A_{UC} 90	A_{UF} 90	A_{RW} 75	S_F 50	M_F 65	M_L 75	C_L 60
5.1	**Bipolar Technology**															
	Gate/Logic Arrays, Digital (Ea = .4)															
	1 - 100 Gates	(16 Pin DIP)	.0036	.012	.024	.024	.035	.025	.030	.032	.049	.047	.0036	.030	.069	1.2
	101 - 1000 Gates	(24 Pin DIP)	.0060	.020	.038	.037	.055	.039	.048	.051	.077	.074	.0060	.046	.11	1.9
	1001 to 3000 Gates	(40 Pin DIP)	.011	.035	.066	.065	.097	.070	.085	.091	.14	.13	.011	.082	.19	3.3
	3001 to 10,000 Gates	(128 Pin PGA)	.033	.12	.22	.22	.33	.23	.28	.30	.46	.44	.033	.28	.65	12
	10,000 to 30,000 Gates	(180 Pin PGA)	.052	.17	.33	.33	.48	.34	.42	.45	.68	.65	.052	.41	.95	17
	30,000 to 60,000 Gates	(224 Pin PGA)	.075	.23	.44	.43	.63	.46	.56	.61	.90	.85	.075	.53	1.2	21
5.1	Linear Microcircuits (Ea = .65)															
	1 - 100 Transistors	(14 Pin DIP)	.0095	.024	.039	.034	.049	.057	.062	.12	.13	.076	.0095	.044	.096	1.1
	101 - 300 Transistors	(18 Pin DIP)	.017	.041	.065	.054	.078	.10	.11	.22	.24	.13	.017	.072	.15	1.4
	301 - 1000 Transistors	(24 Pin DIP)	.033	.074	.11	.092	.13	.19	.19	.41	.44	.22	.033	.12	.26	2.0
	1001 - 10,000 Transistors	(40 Pin DIP)	.050	.12	.18	.15	.21	.29	.30	.63	.67	.35	.050	.19	.41	3.4
5.1	Programmable Logic Arrays (Ea = .4)															
	Up to 200 Gates	(16 Pin DIP)	.0061	.016	.029	.027	.040	.032	.037	.044	.061	.054	.0061	.034	.076	1.2
	201 to 1000 Gates	(24 Pin DIP)	.011	.028	.048	.045	.065	.054	.063	.077	.10	.089	.011	.057	.12	1.9
	1001 to 5000 Gates	(40 Pin DIP)	.022	.052	.087	.082	.12	.099	.11	.14	.19	.16	.022	.10	.22	3.3
5.1	**MOS Technology**															
	Gate/Logic Arrays, Digital (Ea = .35)															
	1 to 100 Gates	(16 Pin DIP)	.0057	.015	.027	.027	.039	.029	.035	.039	.056	.052	.0057	.033	.074	1.2
	101 to 1000 Gates	(24 Pin DIP)	.010	.028	.045	.043	.062	.049	.057	.066	.092	.083	.010	.053	.12	1.9
	1001 to 3000 Gates	(40 Pin DIP)	.019	.047	.080	.077	.11	.088	.10	.12	.17	.15	.019	.095	.21	3.3
	3001 to 10,000 Gates	(128 Pin PGA)	.049	.14	.25	.24	.36	.27	.32	.36	.51	.48	.049	.30	.69	12
	10,001 to 30,000 Gates	(180 Pin PGA)	.084	.22	.39	.37	.54	.42	.49	.58	.79	.72	.084	.46	1.0	17
	30,000 to 60,000 Gates	(224 Pin PGA)	.13	.31	.53	.51	.73	.59	.69	.82	1.1	.98	.13	.63	1.4	21
5.1	Linear Microcircuits (Ea = .65)															
	1 to 100 Transistors	(14 Pin DIP)	.0095	.024	.039	.034	.049	.057	.062	.12	.13	.076	.0095	.044	.096	1.1
	101 to 300 Transistors	(18 Pin DIP)	.017	.041	.065	.054	.078	.10	.11	.22	.24	.13	.017	.072	.15	1.4
	301 to 1,000 Transistors	(24 Pin DIP)	.033	.074	.11	.082	.13	.19	.19	.41	.44	.22	.033	.12	.26	2.0
	1001 to 10,000 Transistors	(40 Pin DIP)	.05	.12	.18	.15	.21	.29	.30	.63	.67	.35	.05	.19	.41	3.4
5.1	Floating Gate Programmable															
	Logic Array, MOS (Ea = .35)															
	Up to 500 Gates	(24 Pin DIP)	.0046	.018	.035	.035	.052	.035	.044	.044	.070	.070	.0046	.044	.10	1.9
	501 - 2000 Gates	(28 Pin DIP)	.0056	.021	.042	.042	.062	.042	.052	.053	.084	.083	.0056	.052	.12	2.3
	2001 - 5000 Gates	(28 Pin DIP)	.0061	.022	.043	.042	.063	.043	.054	.055	.086	.084	.0061	.053	.13	2.3
	5001 to 20000 Gates	(40 Pin DIP)	.0095	.033	.064	.063	.094	.065	.080	.083	.13	.13	.0095	.079	.19	3.3
5.1	Microprocessors, Bipolar (Ea = .4)															
	Up to 8 Bits	(40 Pin DIP)	.028	.061	.098	.091	.13	.12	.13	.17	.22	.18	.028	.11	.24	3.3
	Up to 16 Bits	(64 Pin PGA)	.052	.11	.18	.16	.23	.21	.24	.32	.39	.31	.052	.20	.41	5.8
	Up to 32 Bits	(128 Pin PGA)	.11	.23	.36	.33	.47	.44	.49	.65	.81	.65	.11	.42	.86	12
5.1	Microprocessors, MOS (Ea = .35)															
	Up to 8 Bits	(40 Pin DIP)	.048	.089	.13	.12	.16	.16	.17	.24	.28	.22	.048	.15	.28	3.4
	Up to 16 Bits	(64 Pin PGA)	.093	.17	.24	.22	.29	.30	.32	.45	.52	.40	.093	.27	.50	5.6
	Up to 32 Bits	(128 Pin PGA)	.19	.34	.49	.45	.60	.61	.66	.90	1.1	.82	.19	.54	1.0	12

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Generic Failure Rate, λ_g (Failures/10^6 Hours) for Microcircuits. See Page A-4 for π_Q Values

(Defaults: π_T Based on Ea Shown, Solder or Weld Seal DIPs/PGAs (No. Pins as Shown Below), π_L = 1 (Device in Production ≥ 2 Yr.))

Section #	Part Type	Environ.→ T_J (°C) →	G_B 50	G_F 60	G_M 65	N_S 60	N_U 65	A_{IC} 75	A_{IF} 75	A_{UC} 90	A_{UF} 90	A_{RW} 75	S_F 50	M_F 65	M_L 75	C_L 60
5.2	**MOS Technology**															
	Memories, ROM (Ea = .6)															
	Up to 16K	(24 Pin DIP)	.0047	.018	.036	.035	.053	.037	.045	.048	.074	.071	.0047	.044	.11	1.9
	16K to 64K	(28 Pin DIP)	.0059	.022	.043	.042	.063	.045	.055	.060	.090	.086	.0059	.053	.13	2.3
	64K to 256K	(28 Pin DIP)	.0067	.023	.045	.044	.068	.048	.059	.068	.099	.089	.0067	.055	.13	2.3
	256K to 1 MB	(40 Pin DIP)	.011	.036	.068	.066	.098	.075	.090	.11	.15	.14	.011	.083	.20	3.3
5.2	Memories, PROM, UVEPROM, EEPROM, EAPROM (Ea = .6) (NOTE λ_{cyc} = 0 Assumed for EEPROM)															
	Up to 16K	(24 Pin DIP)	.0049	.018	.036	.036	.053	.037	.046	.049	.075	.072	.0048	.045	.11	1.9
	16K to 64K	(28 Pin DIP)	.0061	.022	.044	.043	.064	.046	.056	.062	.093	.087	.0062	.054	.13	2.3
	64K to 256K	(28 Pin DIP)	.0072	.024	.046	.045	.067	.051	.061	.073	.10	.092	.0072	.057	.13	2.3
	256K to 1 MB	(40 Pin DIP)	.012	.038	.071	.068	.10	.080	.095	.12	.16	.14	.012	.086	.20	3.3
5.2	Memories, DRAM (Ea = .6)															
	Up to 16K	(18 Pin DIP)	.0040	.014	.027	.027	.040	.029	.035	.040	.059	.055	.0040	.034	.080	1.4
	16K to 64K	(22 Pin DIP)	.0055	.019	.036	.034	.051	.039	.047	.056	.079	.070	.0055	.043	.10	1.7
	64K to 256K	(24 Pin DIP)	.0074	.023	.043	.040	.060	.049	.058	.076	.10	.084	.0074	.051	.12	1.9
	256K to 1 MB	(28 Pin DIP)	.011	.032	.057	.053	.077	.070	.080	.12	.15	.11	.011	.067	.15	2.3
5.2	Memories, SRAM, (MOS & BiMOS) (Ea = .6)															
	Up to 16K	(18 Pin DIP)	.0079	.022	.038	.034	.050	.048	.054	.083	.10	.073	.0079	.044	.098	1.4
	16K to 64K	(22 Pin DIP)	.014	.034	.057	.050	.073	.077	.085	.14	.17	.11	.014	.065	.14	1.8
	64K to 256K	(24 Pin DIP)	.023	.053	.084	.071	.10	.12	.13	.25	.27	.16	.023	.092	.19	1.9
	256K to 1 MB	(28 Pin DIP)	.043	.092	.14	.11	.16	.22	.23	.46	.49	.26	.043	.15	.30	2.3
5.2	**Bipolar Technology**															
	Memories, ROM, PROM (Ea = .6)															
	Up to 16K	(24 Pin DIP)	.010	.028	.050	.046	.067	.062	.070	.10	.13	.096	.010	.058	.13	1.9
	16K to 64K	(28 Pin DIP)	.017	.043	.071	.063	.091	.095	.11	.18	.21	.14	.017	.081	.18	2.3
	64K to 256K	(28 Pin DIP)	.028	.065	.10	.085	.12	.15	.16	.30	.33	.19	.028	.11	.23	2.3
	256K to 1 MB	(40 Pin DIP)	.053	.12	.18	.15	.21	.27	.29	.56	.61	.33	.053	.19	.39	3.4
5.2	Memories, SRAM (Ea = .6)															
	Up to 16K	(24 Pin DIP)	.0075	.023	.043	.041	.060	.050	.058	.077	.10	.084	.0075	.052	.12	1.9
	16K to 64K	(28 Pin DIP)	.012	.033	.058	.054	.079	.072	.083	.12	.15	.11	.012	.069	.15	2.3
	64K to 256K	(28 Pin DIP)	.018	.045	.074	.065	.095	.10	.11	.19	.22	.14	.018	.084	.18	2.3
	256K to 1 MB	(40 Pin DIP)	.033	.079	.13	.11	.16	.18	.20	.35	.39	.24	.033	.14	.30	3.4
5.3	VHSIC Microcircuits, CMOS	Refer to Section 5.3, VHSIC CMOS														
5.4	GaAs MMIC (Ea = 1.5)															
	1 to 100 Elements	(8 Pin DIP)	.0013	.0052	.010	.010	.016	.011	.013	.015	.022	.021	.0013	.013	.031	.57
	101 to 1000 Active Elements (Default: Driver and High Power (> 100 mW))	(16 Pin DIP)	.0028	.011	.022	.022	.034	.023	.028	.030	.047	.045	.0028	.028	.068	1.2
5.4	GaAs Digital (Ea = 1.4)															
	1 to 1000 Active Elements	(36 Pin DIP)	.0066	.026	.052	.052	.078	.054	.067	.078	.12	.11	.0066	.065	.16	2.9
	1001 to 10,000 Active Elements	(64 Pin PGA)	.013	.050	.10	.10	.15	.10	.13	.15	.23	.20	.013	.13	.30	5.5

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Quality Factors - π_Q

Description	π_Q
Class S Categories: 1. Procured in full accordance with MIL-M-38510, Class S requirements. 2. Procured in full accordance with MIL-I-38535 and Appendix B thereto (Class U). 3. Hybrids: (Procured to Class S requirements (Quality Level K) of MIL-H-38534.	.25
Class B Categories: 1. Procured in full accordance with MIL-M-38510, Class B requirements. 2. Procured in full accordance with MIL-I-38535, (Class Q). 3. Hybrids: Procured to Class B requirements (Quality Level H) of MIL-H-38534.	1.0
Class B-1 Category: Fully compliant with all requirements of paragraph 1.2.1 of MIL-STD-883 and procured to a MIL drawing, DESC drawing or other government approved documentation. (Does not include hybrids). For hybrids use custom screening section below.	2.0

Learning Factor - π_L

Years in Production, Y	π_L
≤ .1	2.0
.5	1.8
1.0	1.5
1.5	1.2
≥ 2.0	1.0

$\pi_L = .01 \exp(5.35 - .35Y)$

Y = Years generic device type has been in production

Quality Factors (cont'd): π_Q Calculation for Custom Screening Programs

Group	MIL-STD-883 Screen/Test (Note 3)	Point Valuation
1*	TM 1010 (Temperature Cycle, Cond B Minimum) and TM 2001 (Constant Acceleration, Cond B Minimum) and TM 5004 (or 5008 for Hybrids) (Final Electricals @ Temp Extremes) and TM 1014 (Seal Test, Cond A, B, or C) and TM 2009 (External Visual)	50
2*	TM 1010 (Temperature Cycle, Cond B Minimum) or TM 2001 (Constant Acceleration, Cond B Minimum) TM 5004 (or 5008 for Hybrids) (Final Electricals @ Temp Extremes) and TM 1014 (Seal Test, Cond A, B, or C) and TM 2009 (External Visual)	37
3	Pre-Burn In Electricals TM 1015 (Burn-In B-Level/S-Level) and TM 5004 (or 5008 for Hybrids) (Post Burn-In Electricals @ Temp Extremes)	30 (B Level) 36 (S Level)
4*	TM 2020 Pind (Particle Impact Noise Detection)	11
5	TM 5004 (or 5008 for Hybrids) (Final Electricals @ Temperature Extremes)	11 (Note 1)
6	TM 2010/17 (Internal Visual)	7
7*	TM 1014 (Seal Test, Cond A, B, or C)	7 (Note 2)
8	TM 2012 (Radiography)	7
9	TM 2009 (External Visual)	7 (Note 2)
10	TM 5007/5013 (GaAs) (Wafer Acceptance)	1
11	TM 2023 (Non-Destructive Bond Pull)	1

$$\pi_Q = 2 + \frac{87}{\Sigma \text{ Point Valuations}}$$

*NOT APPROPRIATE FOR PLASTIC PARTS.

NOTES:
1. Point valuation only assigned if used independent of Groups 1, 2 or 3.
2. Point valuation only assigned if used independent of Groups 1 or 2.
3. Sequencing of tests within groups 1, 2 and 3 must be followed.
4. TM refers to the MIL-STD-883 Test Method.
5. Nonhermetic parts should be used only in controlled environments (i.e., G_B and other temperature/humidity controlled environments).

EXAMPLES:

1. Mfg. performs Group 1 test and Class B burn-in: $\pi_Q = 2 + \frac{87}{50+30} = 3.1$
2. Mfg. performs internal visual test, seal test and final electrical test: $\pi_Q = 2 + \frac{87}{7+7+11} = 5.5$

Other Commercial or Unknown Screening Levels	$\pi_Q = 10$

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Generic Failure Rate - λ_g (Failures/10^6 Hours) for Discrete Semiconductors

Section #	Part Type	Env.→ G_B	G_F	G_M	N_S	N_U	A_{IC}	A_{IF}	A_{UC}	A_{UF}	A_{RW}	S_F	M_F	M_L	C_L
		T_J(°C) → 50	60	65	60	65	75	75	90	90	75	50	65	75	60
	DIODES														
6.1	General Purpose Analog	.0036	.028	.049	.043	.10	.092	.21	.20	.44	.17	.0018	.076	.23	1.5
6.1	Switching	.00094	.0075	.013	.011	.027	.024	.054	.054	.12	.045	.00047	.020	.060	.40
6.1	Fast Recovery Pwr. Rectifier	.023	.19	.32	.28	.68	.61	1.4	1.3	2.9	1.1	.012	.50	1.5	10
6.1	Power Rectifier/ Schottky Pwr.	.0028	.022	.039	.034	.082	.073	.16	.16	.35	.13	.0014	.060	.18	1.2
6.1	Transient Suppressor/Varistor	.0029	.023	.040	.035	.084	.075	.17	.17	.36	.14	.0015	.062	.18	1.2
6.1	Voltage Ref/Reg. (Avalanche and Zener)	.0033	.024	.039	.035	.082	.066	.15	.13	.27	.12	.0016	.060	.16	1.3
6.1	Current Regulator	.0056	.040	.066	.060	.14	.11	.25	.22	.46	.21	.0028	.10	.28	2.1
6.2	Si Impatt (f ≤ 35 GHz)	.86	2.8	8.9	5.6	20	11	14	36	62	44	.43	16	67	350
6.2	Gunn/Bulk Effect	.31	.76	2.1	1.5	4.6	2.0	2.5	4.5	7.6	7.9	.16	3.7	12	94
6.2	Tunnel and Back	.004	.0096	.027	.019	.058	.025	.032	.057	.097	.10	.002	.048	.15	1.2
6.2	PIN	.028	.068	.19	.14	.41	.18	.22	.40	.69	.71	.014	.34	1.1	8.5
6.2	Schottky Barrier and Point Contact (200 MHz ≤ f ≤ 35 GHz)	.047	.11	.31	.23	.68	.30	.37	.67	1.1	1.2	.023	.56	1.8	14
6.2	Varactor	.012	.026	.072	.052	.16	.069	.086	.15	.26	.28	.0054	.13	.41	3.3
6.10	Thyristor/SCR	.0025	.020	.034	.030	.072	.064	.14	.14	.31	.12	.0012	.053	.16	1.1
	TRANSISTORS														
6.3	NPN/PNP (f < 200 MHz)	.00015	.0011	.0017	.0017	.0037	.0030	.0067	.0060	.013	.0056	.000073	.0027	.0074	.056
6.3	Power NPN/PNP (f < 200 MHz)	.0057	.042	.069	.063	.15	.12	.26	.23	.50	.22	.0029	.11	.29	2.2
6.4	Si FET (f ≤ 400 MHz)	.014	.099	.16	.15	.34	.28	.62	.53	1.1	.51	.0069	.25	.68	5.3
6.9	Si FET (f > 400 MHz)	.099	.24	.64	.47	1.4	.61	.76	1.3	2.3	2.4	.049	1.2	3.6	30
6.8	GaAs FET (P < 100 mW)	.17	.51	1.5	1.0	3.4	1.8	2.3	5.4	9.2	7.2	.083	2.8	11	63
6.8	GaAs FET (P ≥ 100 mW)	.42	1.3	3.9	2.5	8.5	4.5	5.6	13	23	18	.21	6.9	27	160
6.5	Unijunction	.016	.12	.20	.18	.42	.36	.80	.74	1.6	.66	.0079	.31	.88	6.4
6.6	RF, Low Noise (f > 200 MHz, P < 1W)	.094	.23	.63	.46	1.4	.60	.75	1.3	2.3	2.4	.047	1.1	3.6	28
6.7	RF, Power (P ≥ 1W)	.045	.091	.23	.18	.50	.18	.23	.32	.55	.73	.023	.41	1.1	11

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APPENDIX A: PARTS COUNT

Generic Failure Rate - λ_g (Failures/10^6 Hours) for Discrete Semiconductors (cont'd)

Section #	Part Type	Env.→ G_B	G_F	G_{MI}	N_S	N_U	A_{IC}	A_{IF}	A_{UC}	A_{UF}	A_{FAW}	S_F	M_F	M_L	C_L
	T_J(°C)→ 50	60	65	60	65	75	75	90	90	75	50	65	75	60
	OPTO-ELECTRONICS														
6.11	Photodetector	.011	.029	.13	.074	.20	.084	.13	.17	.23	.36	.0057	.15	.51	6.6
		.027	.070	.31	.17	.47	.20	.30	.42	.56	.85	.013	.35	1.2	16
6.11	Opto-Isolator	.00047	.0012	.0056	.0031	.0084	.0035	.0053	.0074	.0098	.015	.00024	.0063	.021	28
6.11	Emitter	.0062	.016	.073	.040	.11	.046	.059	.096	.13	.20	.0031	.082	.28	3.6
6.12	Alphanumeric Display	5.1	16	78	39	120	58	86	86	110	240	2.6	87	350	3600
6.13	Laser Diode, GaAs/Al GaAs	9.0	28	135	69	200	100	150	150	200	400	4.5	150	600	6200
6.13	Laser Diode, In GaAs/In GaAsP	See Section 7 (Includes Receivers, CRTs, Cross Field Amplifiers, Klystrons, TWTs, Magnetrons)													
7	TUBES	See Section 8													
8	LASERS														

Discrete Semiconductor Quality Factors - π_Q

Section Number	Part Types	JANTXV	JANTX	JAN	Lower	Plastic
6.1, 6.3, 6.4, 6.5, 6.10, 6.11, 6.12	Non-RF Devices/ Opto-Electronics*	.70	1.0	2.4	5.5	8.0
6.2	High Freq Diodes	.50	1.0	5.0	25	50
6.2	Schottky Diodes	.50	1.0	1.8	2.5	----
6.6, 6.7, 6.8, 6.9	RF Transistors	.50	1.0	2.0	5.0	----
6.13	*Laser Diodes	π_Q = 1.0 Hermetic Package = 1.0 Nonhermetic with Facet Coating = 3.3 Nonhermetic without Facet Coating				

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Generic Failure Rate, λ_g (Failure/10^6 Hours) For Resistors (Section 9.1)

Part Type	Style	MIL-R-	Env. → G_B T_A(°C) → 30	Q_F 40	G_M 45	N_S 40	N_U 45	A_{IC} 55	A_{IF} 55	A_{UC} 70	A_{UF} 70	A_{RW} 55	S_F 30	M_F 45	M_L 55	C_L 40
Composition	RCR	39008	.0022	.011	.051	.034	.13	.071	.091	.17	.23	.25	.0011	.12	.34	4.9
Composition	RC	11	.0022	.011	.051	.034	.13	.071	.091	.17	.23	.25	.0011	.12	.34	4.9
Film, Insulated	RLR	39017	.0037	.016	.07	.05	.18	.08	.11	.16	.22	.29	.0018	.16	.40	7.0
Film, Insulated	RL	22684	.0037	.016	.07	.05	.18	.08	.11	.16	.22	.29	.0018	.16	.40	7.0
Film, RN (R, C or N)	RNR	55182	.0037	.016	.07	.05	.18	.08	.11	.16	.22	.29	.0018	.16	.40	7.0
Film, Chip	RM	55342	.0037	.016	.07	.05	.18	.08	.11	.16	.22	.29	.0018	.16	.40	7.0
Film	RN	10509	.0037	.016	.07	.05	.18	.08	.11	.16	.22	.29	.0018	.16	.40	7.0
Film, Power	RD	11804	.010	.041	.16	.12	.43	.18	.24	.32	.44	.65	.0051	.38	.89	18
Film, Network	RZ	83401	.0016	.0084	.038	.025	.10	.053	.068	.12	.17	.19	.00082	.088	.26	3.6
Wirewound, Accurate	RBR	39005	.0024	.010	.044	.031	.11	.054	.069	.11	.15	.19	.0012	.10	.26	4.5
Wirewound, Accurate	RB	93	.0024	.010	.044	.031	.11	.054	.069	.11	.15	.19	.0012	.10	.26	4.5
Wirewound, Power	RWR	39007	.0085	.038	.16	.11	.41	.19	.25	.38	.52	.68	.0043	.36	.94	16
Wirewound, Power	RW	26	.0085	.038	.16	.11	.41	.19	.25	.38	.52	.68	.0043	.36	.94	16
Wirewound, Power, Chassis Mounted	RER	39009	.016	.070	.29	.21	.77	.36	.46	.71	.98	1.3	.0080	.68	1.8	30
Wirewound, Power, Chassis Mounted	RE	18546	.016	.070	.29	.21	.77	.36	.46	.71	.98	1.3	.0080	.68	1.8	30
Thermistor	RTH	23648	.0014	.0058	.023	.017	.061	.026	.033	.045	.062	.091	.0007	.054	.13	2.5
Wirewound, Variable	RTR	39015	.0024	.010	.044	.031	.12	.054	.069	.11	.15	.19	.0012	.10	.26	4.5
Wirewound, Variable	RT	27208	.0024	.010	.044	.031	.12	.054	.069	.11	.15	.19	.0012	.10	.26	4.5
Wirewound, Variable, Precision	RR	12934	.0024	.010	.044	.031	.12	.054	.069	.11	.15	.19	.0012	.10	.26	4.5
Wirewound, Variable, Semiprecision	RA	19	.0026	.013	.059	.037	.15	.083	.11	.19	•	•	.0013	•	•	•
Wirewound, Variable, Semiprecision	RK	39002	.0026	.013	.059	.037	.15	.083	.11	.19	•	•	.0013	•	•	•
Wirewound, Variable, Power	RP	22	.0024	.010	.044	.031	.12	.054	.069	.11	.15	.19	.0012	.10	.26	4.5
Nonwirewound, Variable	RJR	39035	.0037	.016	.068	.048	.18	.083	.11	.16	.22	.29	.0018	.16	.40	7.0
Nonwirewound, Variable	RJ	22097	.0037	.016	.068	.048	.18	.083	.11	.16	.22	.29	.0018	.16	.40	7.0
Composition, Variable	RV	94	.0037	.016	.068	.048	.18	.083	.11	.16	.22	.29	.0018	.16	.40	7.0
Nonwirewound, Variable Precision	RQ	39023	.040	.020	.091	.061	.24	.13	.16	.30	.42	.45	.0020	.21	.62	8.7
Film, Variable	RVC	23285	.040	.020	.091	.061	.24	.13	.16	.30	.42	.45	.0020	.21	.62	8.7

NOTES: 1) * Not Normally used in this Environment
2) T_A = Default Component Ambient Temperature (°C)
3) Default Pwr. dissipation .5 watts assumed for all categories except RD, RWR, RW, RER and RE styles. RD, RWR, RW: 8 watts. RER and RE: 40 watts.

Quality	Established Reliability Styles			MIL-SPEC	
	S	R	P	M	Lower
π_Q	.030	.10	.30	1.0	3.0	10

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Generic Failure Rate, λ_g (Failures/10^6 Hours) for Capacitors (Section 10.1)

Part Type or Dielectric	Style	MIL-C-	Env.→ G_B T_A (°C)→ 30	G_F 40	G_M 45	N_S 40	N_U 45	A_{IC} 55	A_{IF} 55	A_{UC} 70	A_{UF} 70	A_{RW} 55	S_F 30	M_F 45	M_L 55	C_L 40
Paper, By-Pass	CP	25	.00051	.0061	.013	.0043	.010	.0095	.012	.025	.030	.032	.00025	.013	.039	.35
Paper, By-Pass	CA	12889	.00051	.0061	.013	.0043	.010	.0095	.012	.025	.030	.032	.00025	.013	.039	.35
Paper/Plastic, Feed-through	CZ, CZR	11693	.00051	.0061	.013	.0043	.010	.0095	.012	.025	.030	.032	.00025	.013	.039	.35
Paper/Plastic Film	CQ, CQR	19978	.00070	.0084	.018	.0059	.014	.013	.016	.034	.041	.043	.00035	.018	.054	.48
Metallized Plastic/ Plastic	CH	18312	.00051	.0061	.013	.0043	.010	.0095	.012	.025	.030	.032	.00025	.013	.039	.35
Metallized Paper/Plastic	CHR	39022	.00070	.0084	.018	.0059	.014	.013	.016	.034	.041	.043	.00035	.018	.054	.48
Metallized Paper/Plastic	CFR	55514	.00070	.0084	.018	.0059	.014	.013	.016	.034	.041	.043	.00035	.018	.054	.48
Metallized Plastic	CRH	83421	.00070	.0084	.018	.0059	.014	.013	.016	.034	.041	.043	.00035	.018	.054	.48
MICA (Dipped)	CM	5	.00057	.0088	.022	.0062	.016	.019	.024	.069	.082	.064	.00029	.022	.080	.50
MICA (Dipped or Molded)	CMR	39001	.00057	.0088	.022	.0062	.016	.019	.024	.069	.082	.064	.00029	.022	.080	.50
MICA (Button)	CB	10950	.00057	.0088	.022	.0062	.016	.019	.024	.069	.082	.064	.00029	.022	.080	.50
Glass	CYR	23269	.0010	.016	.039	.011	.029	.034	.043	.12	.15	.11	.00051	.039	.14	.90
Glass	CY	11272	.0010	.016	.039	.011	.029	.034	.043	.12	.15	.11	.00051	.039	.14	.90
Ceramic (Gen. Purpose)	CK	11015	.0017	.026	.064	.018	.048	.057	.071	.20	.24	.19	.00086	.064	.24	1.5
Ceramic (Gen. Purpose)	CKR	39014	.0017	.026	.064	.018	.048	.057	.071	.20	.24	.19	.00086	.064	.24	1.5
Ceramic (Temp. Comp.)	CC, CCR	20	.0017	.026	.064	.018	.048	.057	.071	.20	.24	.19	.00086	.064	.24	1.5
Ceramic Chip	CDR	55681	.0035	.053	.13	.037	.098	.12	.14	.41	.49	.38	.0017	.13	.48	3.0
Tantalum, Solid	CSR	39003	.0014	.017	.037	.012	.027	.026	.032	.068	.082	.087	.00070	.037	.11	.96
Tantalum, Chip	CWR	55365	.00014	.0016	.0036	.0011	.0027	.0025	.0031	.0066	.0079	.0084	.000068	.0036	.010	.093
Tantalum, Non-Solid	CLR	39006	.0022	.026	.057	.018	.042	.040	.050	.11	.13	.13	.0011	.057	.17	1.5
Tantalum, Non-Solid	CL	3965	.0022	.026	.057	.018	.042	.040	.050	.11	.13	.13	.0011	.057	.17	1.5
Tantalum, Non-Solid	CRL	83500	.0022	.026	.057	.018	.042	.040	.050	.11	.13	.13	.0011	.057	.17	1.5
Aluminum Oxide	CU, CUR	39018	.0013	.019	.047	.014	.036	.042	.052	.15	.18	.14	.00063	.047	.17	1.1
Aluminum Dry	CE	62	.0013	.019	.047	.014	.036	.042	.052	.15	.18	.14	.00063	.047	.17	1.1
Variable, Ceramic	CV	81	.0055	.066	.14	.046	.11	.10	.13	.27	.32	.34	.0027	.14	.42	3.8
Variable, Piston	PC	14409	.0047	.073	.18	.051	.13	.16	.20	.57	.68	.53	.0024	.18	.66	4.1
Variable, Air Trimmer	CT	92	.0000057	.000087	.00021	.000061	.00016	.00019	.00024	.00068	.00081	.00063	.0000028	.00021	.00079	.0050
Variable, Vacuum	CG	23183	.0042	.050	.11	.035	.062	.077	.097	.20	.24	.26	.0021	.11	.32	2.9

NOTES:
1) * Not Normally used in this Environment
2) T_A = Default Component Ambient Temperature (°C)
3) Voltage stress = .4, π_{SR} = 1
4) Assumed capacitance (µF): CP, CA, CZ, CZR, CQ, CQR, CH, CHR, CFR, CRH: 3.0; CM, CMR, CB: 0.003; CYR, CY, CK, CKR, CC, CCR, CDR: 20; CSR: 150; CWR: 50; CLR, CL, CRL: 1000; CU, CUR, CE: 6000; CV, PC, CT, CG: 0.00006

Quality	Established Reliability Styles D	C	S, B	R	P	M	L	MIL-SPEC	Lower
π_Q	.001	.01	.030	.10	.30	1.0	1.5	3.0	10

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Generic Failure Rate, λ_g (Failures/10^6 Hours) for Inductive, Electromechanical and Miscellaneous Parts

Section #	Part Type	MIL-	Env.→ G_B T_A (°C)→30	G_F 40	G_M 45	N_S 40	N_U 45	A_{IC} 55	A_{IF} 55	A_{UC} 70	A_{UF} 70	A_{RW} 55	S_F 30	M_F 45	M_L 55	C_L 40
	INDUCTIVE DEVICES															
11.1	Transformer, Switching	T-21038	.00061	.0042	.0090	.0035	.012	.0051	.0067	.0070	.0080	.020	.00031	.0097	.029	.43
11.1	Transformer, Flyback	T-27	.0058	.040	.085	.033	.11	.048	.064	.066	.085	.19	.0029	.092	.27	4.0
11.1	Transformer, Audio	T-27	.015	.10	.22	.086	.29	.12	.17	.17	.22	.50	.0075	.24	.70	10
11.1	Transformer, Power	T-27	.053	.36	.77	.30	1.0	.44	.58	.60	.77	1.7	.026	.83	2.5	37
11.1	Transformer, RF	T-55631	.14	.96	2.0	.80	2.7	1.2	1.5	1.6	2.1	4.6	.070	2.2	6.5	97
11.2	Coil, Fixed Inductor or Choke	C-15305, C-39010	.000032	.00022	.00047	.00018	.00063	.00027	.00036	.00037	.00047	.0011	.00002	.00051	0.0015	.022
11.2	Coil, Variable Inductor	C-15305	.00005	.00037	.00079	.00031	.0010	.00044	.00059	.00061	.00079	.0018	.00003	.00085	.0025	.037
	ROTATING DEVICES															
12.1	Motors, General		6.9	6.8	8.3	6.8	8.3	13	13	30	30	13	6.9	.	.	.
12.1	Sensor Motor		27	27	33	27	33	52	52	1.2e+02	1.2e+02	52	27	33	52	27
12.1	Servo Motor		5.4	5.4	6.5	5.4	6.5	10	10	23	23	10	5.4	6.5	10	5.4
12.1	Stepper Motor		1.2	1.2	1.4	1.2	1.4	2.3	2.3	5.3	5.3	2.3	1.2	1.4	2.3	1.2
12.2	Synchros		.031	.071	.47	.25	.70	.19	.28	1.1	1.8	1.2	.016	.54	1.7	24
12.2	Resolvers		.047	.11	.70	.37	1.0	.28	.43	1.7	2.6	1.8	.023	.81	2.6	.
	ELAPSED TIME METERS															
12.3	ETM-AC		10	20	120	70	180	50	80	160	250	260	5.0	140	380	.
12.3	ETM-Inverter Driver		15	30	180	105	270	75	120	240	375	390	7.5	210	570	.
13.3	ETM-Commutator DC		40	80	480	280	720	200	320	640	1000	1040	20	560	1520	.
	RELAYS															
13.1	General Purpose (Bal. Arm.)		.049	.12	1.0	.50	1.9	.60	.77	1.3	1.4	3.9	.025	1.7	5.7	.
13.1	Sensitive (Bal. Arm.)		.099	.25	2.1	.98	3.7	1.2	1.5	2.5	2.8	7.9	.049	3.5	11	.
13.1	Dry Reed		.059	.15	1.2	.60	2.2	.72	.93	1.5	1.7	4.7	.030	2.1	6.8	.
13.1	Thermal Bi-metal		.099	.25	2.1	.98	3.7	1.2	1.5	2.5	2.8	7.9	.049	3.5	11	.
13.1	Magnetic Latching, (Bal. Arm.)		.049	.12	1.0	.50	1.9	.60	.77	1.3	1.4	3.9	.025	1.7	5.7	.
13.1	Contactor, High Current (Solenoid)		.049	.12	1.0	.50	1.9	.60	.77	1.3	1.4	3.9	.025	1.7	5.7	.
13.2	Solid State, All		.029	.087	.35	.17	.49	.35	.55	.61	.93	.67	.012	.35	.98	17
	SWITCHES	See 14.1														
14.1	Dual In-line Package		.00012	.00036	.0022	.00096	.0035	.0012	.0022	.0016	.0026	.0055	.00006	.0030	.0080	.14
14.1	Limit		4.3	13	77	34	1.2e+02	43	77	58	95	2.0e+02	2.2	1.1e+02	2.9e+02	5.2e+03
14.1	Microwave		1.7	5.1	31	14	49	17	31	22	37	78	.85	43	1.1e+02	2.0e+03
14.1	Pushbutton		.10	.30	1.8	.80	2.9	1.0	1.8	1.3	2.2	4.6	.050	2.5	6.7	1.2e+02
14.1	Reed		.0010	.0030	.018	.0080	.029	.0100	.018	.013	.022	.046	.00050	.025	.067	1.2
14.1	Rocker		.023	.069	.41	.18	.67	.23	.41	.30	.51	1.1	.012	.57	1.5	28
14.1	Rotary		.11	.33	2.0	.88	3.2	1.1	2.0	1.4	2.4	5.1	.055	2.8	7.4	1.3e+02
14.1	Sensitive		.49	1.5	8.8	3.9	14	4.9	8.8	6.4	11	23	.25	12	33	5.9e+02
14.1	Thermal		.031	.093	.56	.25	.90	.31	.56	.40	.68	1.4	.015	.77	2.1	37
14.1	Thumbwheel		.18	.54	3.2	1.4	5.2	1.8	3.2	2.3	4.0	8.3	.090	4.5	12	2.2e+02
14.1	Toggle		.10	.30	1.8	.80	2.9	1.0	1.8	1.3	2.2	4.6	.050	2.5	6.7	1.2e+02
14.2	Circuit Breaker, All		.68	1.4	10	5.4	18	4.8	6.1	7.5	8.2	31	.34	17	45	.
	CONNECTORS															
15.1	Circular		.0011	.0013	.011	.0065	.018	.0049	.0082	.016	.025	.031	.00055	.014	.044	.64
15.1	PCB Card Edge		.044	.052	.45	.26	.73	.20	.33	.65	.98	1.3	.022	.56	1.8	25
15.1	Hexagonal		.18	.19	1.7	.97	2.7	.74	1.2	2.5	3.7	4.7	.082	2.1	6.7	95
15.1	Rack and Panel		.023	.027	.24	.14	.38	.10	.17	.34	.52	.66	.011	.30	.93	13
15.1	Rectangular		.050	.060	.52	.30	.84	.23	.38	.75	1.1	1.4	.025	.65	2.0	29
15.1	RF Coaxial		.00045	.00053	.0046	.0027	.0075	.0020	.0034	.0067	.0100	.013	.00022	.0058	.018	.26
15.1	Telephone		.0082	.0097	.085	.049	.14	.037	.062	.12	.18	.23	.0041	.11	.33	4.8
15.2	IC Sockets (DIP, SIP, PGA)		.0035	.011	.049	.021	.063	.028	.042	.039	.046	.088	.0018	.049	.13	2.3

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Generic Failure Rate, λ_g (Failures/10^6 Hours) for Inductive, Electromechanical and Miscellaneous Parts

Section #	Part Type	MIL-	Env.→ G_B / T_A (°C)→ 30	G_F 40	G_M 45	N_S 40	N_U 45	A_{IC} 55	A_{IF} 55	A_{UC} 70	A_{UF} 70	A_{RW} 55	S_F 30	M_F 45	M_L 55	C_L 40
16.1	Plated Through Hole Circuit Boards		.022	.045	.16	.11	.29	.11	.18	.36	.62	.42	.011	.22	.60	11
16.2	Surface Mount Tech. Circuit Boards		.0025	.37	1.8	1.8	42	6.1	6.1	35	35	6.1	.0025	11	.11	.11
	SINGLE CONNECTIONS															
17.1	Hand Solder, w/o Wrapping		.0013	.0026	.0091	.0052	.014	.0052	.0078	.0078	.0100	.021	.00065	.012	.031	.55
17.1	Hand Solder, w/Wrapping		7.0e-05	.00014	.00049	.00028	.00077	.00028	.00042	.00042	.00056	.0011	3.5e-05	.00063	.0017	.029
*17.1	Crimp		.00026	.00052	.0018	.0010	.0029	.0010	.0016	.0016	.0021	.0042	.00013	.0023	.0062	.11
17.1	Weld		1.5e-05	3.0e-05	1.0e-04	6.0e-05	0.00017	6.0e-05	9.0e-05	9.0e-05	.00012	.00024	7.5e-06	.00013	.00036	.0063
17.1	Solderless Wrap		6.8e-06	1.4e-05	4.8e-05	2.7e-05	7.5e-05	2.7e-05	4.1e-05	4.1e-05	5.4e-05	.00011	3.4e-06	6.1e-05	.00016	.0029
17.1	Clip Termination		.00012	.00024	.00084	.00048	.0013	.00048	.00072	.00072	.00096	.0019	6.0e-05	.0011	.0029	.050
17.1	Reflow Solder		6.9e-05	.00014	.00048	.00028	.00076	.00028	.00041	.00041	.00055	.0011	3.5e-05	.00062	.0017	.029
17.1	Spring Contact		.17	.34	1.2	.68	1.9	.68	1.0	1.0	1.4	2.7	.085	1.5	4.1	71
17.1	Terminal Block		.062	.12	.43	.25	.68	.25	.37	.37	.50	.99	.031	.56	1.5	26
	METERS, PANEL															
*18.1	DC Ammeter or Voltmeter	M-10304	0.09	0.36	2.3	1.1	3.2	2.5	3.8	5.2	6.6	5.4	0.099	5.4	N/A	N/A
*18.1	AC Ammeter or Voltmeter	M-10304	0.15	0.61	3.8	1.8	5.4	4.3	6.4	8.9	11	9.2	0.17	9.2	N/A	N/A
19.1	Quartz Crystals	C-3098	.032	.096	.32	.19	.51	.38	.54	.70	.90	.74	.016	.42	1.0	16
20.1	Lamps, Incandescent, AC		3.9	7.8	12	12	16	16	16	19	23	19	2.7	16	23	100
20.1	Lamps, Incandescent, DC		13	26	38	38	51	51	51	64	77	64	9.0	51	77	350
	ELECTRONIC FILTERS															
21.1	Ceramic-Ferrite	F-15733	.022	.044	.13	.088	.20	.15	.20	.24	.29	.24	.018	.15	.33	2.6
21.1	Discrete LC Comp.	F-15733	.12	.24	.72	.48	1.1	.84	1.1	1.3	1.6	1.3	.096	.84	1.8	14
21.1	Discrete LC & Crystal Comp.	F-18327	.27	.54	1.6	1.1	2.4	1.9	2.4	3.0	3.5	3.0	.22	1.9	4.1	32
22.1	FUSES		.010	.020	.060	.050	.11	.090	.12	.15	.18	.16	.009	.10	.21	2.3

NOTES:
1) * Not normally used in this environment.
2) T_A = Default Component Ambient Temperature (°C), π_T based on T_A shown.
3) Motor assumptions: 10 yr. (87600 hours) design life assumed; Synchros/Resolvers: Size 10-16, 3 brushes; ETMs: π_T = .5.
4) Relay assumptions: Rated Temp. = 125°C, SPST, Resistive Load, S = .5, 10 cycles/hour.
5) Switch assumptions: SPST; Circuit breakers: DPST, not used as a switch.
6) Connector assumptions: π_K = 1; Sockets: 40 pins.
7) Plated through hole circuit board assumptions: 1000 wave solder joints, 3 planes, no hand soldering; SMT circuit board design assumptions are same as those shown in Section 16.2 example using the default ΔT values shown in Section 16.2.
8) Quartz crystal assumptions: 50 MHz
9) Lamp assumptions: utilization rate = .5, 28 volt rating.

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π_Q Factor for Use with Section 11-22 Devices

Section #	Part Type	Established Reliability	MIL-SPEC	Non-MIL
11.1, 11.2	Inductive Devices	.25*	1.0	3.0
12.1, 12.2, 12.3	Rotating Devices	N/A	N/A	N/A
13.1	Relays, Mechanical	.60	1.5	2.9
13.2	Relays, Solid State and Time Delay (Hybrid & Solid State)	N/A	1.0	1.9
14.1	Switches, Toggle, Pushbutton, Sensitive	N/A	1.0	2.0
14.2	Circuit Breakers	N/A	1.0	8.4
15.1	Connectors	N/A	1.0	2.0
15.2	Connectors, Sockets	N/A	.3	1.0
16.1	Plated Through Hole Circuit Boards	N/A	1.0	2.0
16.2	Surface Mount Tech. Circuit Boards	N/A	N/A	N/A
17.1	Connections	N/A	N/A	N/A
18.1	Meters, Panel	N/A	1.0	3.4
19.1	Quartz Crystals	N/A	1.0	2.1
20.1	Lamps, Incandescent	N/A	N/A	N/A
21.1	Electronic Filters	N/A	1.0	2.9
22.1	Fuses	N/A	N/A	N/A

* Category applies only to MIL-C-39010 Coils.

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APPENDIX A: PARTS COUNT

Default Parameters for Discrete Semiconductors

Section #	Part Type	λ_b	π_T	π_M	π_S	π_C	π_A	π_R	Comments
5.0	MICROCIRCUITS	All Defaults provided with λ_g Table							
	DIODES								
6.1	General Purpose Analog	.0038			.42	1.0			Voltage Stress = .7, Metallurgically Bonded Contacts
6.1	Switching	.001			.42	1.0			Voltage Stress = .7, Metallurgically Bonded Contacts
6.1	Fast Recovery Power Rectifier	.025			.42	1.0			Voltage Stress = .7, Metallurgically Bonded Contacts
6.1	Transient Suppressor/Varistor	.0031			1.0	1.0			Metallurgically Bonded Contacts
6.1	Power Rectifier	.003			.42	1.0			Voltage Stress = .7, Metallurgically Bonded Contacts
6.1	Voltage Ref/Reg. (Avalanche & Zener)	.002			1.0	1.0			Metallurgically Bonded Contacts
6.1	Current Regulator	.0034			1.0	1.0	1.0	1.0	Metallurgically Bonded Contacts
6.2	Si Impatt (≤ 35 GHz)	.22					1.0	1.0	
6.2	Gunn/Bulk Effect	.18					1.0	1.0	
6.2	Tunnel and Back	.0023					1.0	2.0	Rated Power = 1000W
6.2	PIN	.0081					1.0		
6.2	Schottky Barrier and Point Contact (200 MHz ≤ frequency ≤ 35 GHz)	.027			1.0	1.0			
6.2	Varactor	.0025					2.5	1.0	Multiplier Application
6.10	Thyristor/SCR	.0022			.51			1.0	Voltage Stress = .7, Rated Forward Current = 1 Amp
	TRANSISTORS								
6.3	NPN/PNP (f < 200 MHz)	.00074			.21		.70	.77	Voltage Stress = .5, Switching Application, Rated Power = .5W
6.3	Power NPN/PNP (f < 200 MHz)	.00074			.54		1.5	5.5	Voltage Stress = .8, Linear Application, Rated Power = 100W
6.4	Si FET (f ≤ 400 MHz)	.012					.70		MOSFET, Small Signal Switching
6.9	Si FET (f > 400 MHz)	.060							MOSFET
6.8	GaAs FET (P < 100 mW)	.052		1.0			1.0		Low Noise Application, 1 ≤ f ≤ 10 GHz, Input and Output Matching
6.8	GaAs FET (P ≥ 100 mW)	.13		1.0			1.0		Pulsed Application, 5 GHz, 1W Average Output Power, Input and Output Matching
6.5	Unijunction	.0083						.77	
6.6	RF, Low Noise, Bipolar (f > 200 MHz, P < 1W)	.18			.39				Voltage Stress = .7, Rated Power = .5W
6.7	RF, Power (P ≥ 1W)	.08	.36	1.0			1.6		1 GHz, 100W, T_J = 130°C for all Environments, Voltage Stress = .45, Gold Metallization, Pulsed Application, 20% Duty Factor, Input and Output Matching

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Default Parameters for Discrete Semiconductors

Section #	Part Type	λ_b	π_T	π_M	π_S	π_C	π_A	π_R	Comments
	OPTO-ELECTRONICS								
6.11	Photodetector	.0055							Phototransistor
6.11	Opto-Isolator	.013							Phototransistor, Single Device
6.11	Emitter	.00023							LED
6.12	Alphanumeric Display	.0030							7 Character Segment Display
6.13	Laser Diode, GaAs/Al GaAs	3.23			1.0 (π_p)		.77		For Environments with T_J > 75°C, assume T_J = 75°C, Forward Peak Current = .5 Amps (π_I = .62), Pulsed Application, Duty Cycle = .6, Pr/Ps = .5 (π_P = 1)
6.13	Laser Diode, In/GaAs/In GaAsP	5.65			1.0 (π_p)		.77		For Environments with T_J > 75°C, assume T_J = 75°C, Forward Peak Current = .5 Amps (π_I = .62), Pulsed Application, Duty Cycle = .6, Pr/Ps = .5 (π_P = 1)

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26. "VHSIC Impact on System Reliability," RADC-TR-88-13, AD B122629.

27. "Reliability Assessment of Surface Mount Technology," RADC-TR-88-72, AD A193759.

28. "Reliability Prediction Models for Discrete Semiconductor Devices," RADC-TR-88-97, AD A200529.

This study developed new failure rate prediction models for GaAs Power FETS, Transient Suppressor Diodes, Infrared LEDs, Diode Array Displays and Current Regulator Diodes.

29. "Impact of Fiber Optics on System Reliability and Maintainability," RADC-TR-88-124, AD A201946.

30. "VHSIC/VHSIC Like Reliability Prediction Modeling," RADC-TR-89-171, AD A214601.

This study provides the basis for the VHSIC model appearing in MIL-HDBK-217F, Section 5.

31. "Reliability Assessment Using Finite Element Techniques," RADC-TR-89-281, AD A216907.

This study addresses surface mounted solder interconnections and microwire board's plated-through-hole (PTH) connections. The report gives a detailed account of the factors to be considered when performing an FEA and the procedure used to transfer the results to a reliability figure-of-merit.

32. "Reliability Analysis/Assessment of Advanced Technologies," RADC-TR-90-72, ADA 223647.

This study provides the basis for the revised microcircuit models (except VHSIC and Bubble Memories) appearing in MIL-HDBK-217F, Section 5.

33. "Improved Reliability Prediction Model for Field-Access Magnetic Bubble Devices," AFWAL-TR-81-1052.

34. "Reliability/Design Thermal Applications," MIL-HDBK-251.

35. "NASA Parts Application Handbook," MIL-HDBK-978-B (NASA).
This handbook is a five volume series which discusses a full range of electrical, electronic and electromechanical component parts. It provides extensive detailed technical information for each component part such as: definitions, construction details, operating characteristics, derating, failure mechanisms, screening techniques, standard parts, environmental considerations, and circuit application.

36. "Nonelectronic Parts Reliability Data 1991," NPRD-91.
This report contains field failure rate data on a variety of electrical, mechanical, electromechanical and microwave parts and assemblies (1400 different part types). It is available from the Reliability Analysis Center, PO Box 4700, Rome, NY 13440-8200, Phone: (315) 337-0900.

37. "Reliability Assessment of Critical Electronic Components," RL-TR-92-197, AD-A256996.
This study is the basis for new or revised failure rate models in MIL-HDBK-217F, Notice 2, for the following device categories: resistors, capacitors, transformers, coils, motors, relays, switches, circuit breakers, connectors, printed circuit boards and surface mount technology.

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APPENDIX C: BIBLIOGRAPHY

38. "Handbook of Reliability Prediction Procedures for Mechanical Equipment," NSWC-94/L07. This Handbook includes a methodology for nineteen basic mechanical components for evaluating a design for R&M that considers the material properties, operating environment and critical failure modes. It is available from the Carderock Division, Naval Surface Warfare Center, Bethesda, MD 20084-5000, Phone (301) 227-1694.

Custodians:
Army - CR
Navy - EC
Air Force - 17

Review Activities:
Army - MI, AV, ER
Navy - SH, AS, OS
Air Force - 11, 13, 15, 19, 99

User Activities:
Army - AT, ME, GL
Navy - CG, MC, YD, TD
Air Force - 85

Preparing Activity:
Air Force - 17

Project No. RELI-0074

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instruction - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-HDBK-217F, Notice 2	Reliability Prediction of Electronic Equipment

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

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